

Do We Still Need to Be Agile? What Student Projects Reveal About Coordination Skills in the GenAI Era

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Abstract—Generative AI can make local software implementation faster, but it may also raise the importance of shared architectural intent, integration, and human coordination. This study examines that tension across two cohorts of student software projects comprising 14 teams and three synchronized checkpoints ($T_1/T_2/T_3$). We combine multi-checkpoint surveys, repository mining, independent blind-scored evaluator scorecards, and transcript-based evidence to study Generative AI adoption and role perceptions, the temporal density of repository activity, and associations between upfront architectural artifacts, late rework, and project outcomes. The results show that self-reported AI use is widespread and task-oriented, but perception families do not form a single longitudinal trajectory. Repository activity and clean source/test change concentrate near final integration, but internal contrasts show that this late concentration includes cases consistent with compensatory recovery, cases without evaluator-score gain, and cases where improvement occurs with more distributed activity. Candidate coordination-language density increases across the observed transcript checkpoints, while observable T_1 architectural-artifact scope varies substantially without a simple monotonic association with T_3 clean-rework magnitude, baseline-conditioned rework ratios, or independent outcomes. Robustness checks using non-overlapping phases, operational-regularity proxies, influence diagnostics, complexity profiles, and explicit transcript coverage make the main validity threats inspectable but do not remove the study’s observational limits. These findings suggest that Agile remains necessary, but AI-assisted development requires a more explicit and inspectable layer of architectural planning, interfaces, dependencies, and acceptance conditions. Spec-Driven Development is discussed as a plausible, testable response to the coordination gap exposed by the combined timing, planning, rework, complexity, and influence diagnostics.

I. INTRODUCTION

The Agile Manifesto revolutionized software engineering by prioritizing working software, individual interactions, and responding to change over comprehensive documentation and strict upfront planning. For over two decades, this empirical paradigm was the gold standard for navigating requirements uncertainty, and for good reason: in a world where local implementation effort and changing requirements made heavyweight plans costly, minimizing unnecessary upfront documentation helped reduce waste.

That world is changing. Generative AI assistants now produce substantial amounts of code in seconds, shifting scarce software-engineering effort toward architectural alignment, integration, and cognitive coordination. Under this new cost equation, the Agile bet—that planning can remain deliberately

light as understanding develops—deserves to be tested rather than assumed.

If deferring design in an AI-assisted environment merely relocates effort, late project activity may reflect several mechanisms: compensatory recovery, delayed integration without assessed improvement, distributed progress, or technical-complexity pressure. This paper examines how such mechanisms become visible through repository metadata, multi-checkpoint student surveys, independent evaluator scorecards, and qualitative meeting transcripts across two cohorts of software-engineering projects. The resulting evidence connects what students report, what repositories reveal, what evaluators score, and what coordination language appears in the project record.

We structure the investigation around three questions:

- **RQ1 (Descriptive):** How do student developers utilize Generative AI, and how does this adoption shape their longitudinal perceptions of software engineering roles and project expectations?
- **RQ2 (Descriptive):** How does the temporal density of repository activity contrast with the qualitative typification of human coordination friction across the project lifecycle?
- **RQ3 (Primary):** How does the observable scope and content evidence of upfront architectural artifacts at baseline T_1 associate with late-stage clean-path rework and independent, blind-scored project outcomes?

The study makes three contributions. First, it provides a longitudinal, multi-method account of AI-assisted student development that keeps perceived AI impact, self-reported adoption, repository activity, and qualitative coordination evidence analytically distinct. Second, it makes the temporal contrast between repository activity and integration pressure observable and qualifies that contrast using non-overlapping phase bins, timing/outcome contrasts, operational regularity, and coverage-aware transcript evidence while preserving the different grains and coverage of the underlying evidence. Third, it examines upfront architectural-artifact evidence, late clean-path rework, and independent outcomes without collapsing structural observations into an unsupported quality score. The resulting picture is deliberately more useful than a simple success/failure verdict: rather than eliminating threats such as ordinary deadline pressure, process heterogeneity, small-sample sensitivity, and technical complexity, it makes them

explicit and inspectable. It shows where implementation speed meets the harder problem of shared system intent.

The findings motivate a practical proposition. Generative AI does not remove the need for Agile practice; it increases the importance of making architectural intent, interfaces, dependencies, and acceptance conditions explicit while remaining open to revision. These observations motivate Spec-Driven Development (SDD) as a theoretically grounded, testable hypothesis: carrying a specification through technical planning, task decomposition, implementation, and convergence checks may help teams address the coordination bottlenecks exposed by the combined timing, planning, rework, complexity, and influence diagnostics [1], [2]. This observational study neither implemented SDD nor compared it with another development approach; therefore, it does not establish SDD's effectiveness. Instead, the hypothesis invites future intervention research into how teams can preserve adaptability while ensuring that people and AI systems act on the same inspectable account of what the system is meant to become.

II. BACKGROUND AND RELATED WORK

A. The Agile value hierarchy and coordination risk

The Agile Manifesto prioritizes working software, interaction, and response to change over comprehensive documentation [3]. This emphasis remains valuable when requirements evolve, but it does not imply that shared architectural intent is unnecessary. Classic accounts of software coordination emphasize that communication costs grow with system size and team interaction, especially when structure remains tacit [4], [5]. The relevant question for AI-assisted development is therefore not whether teams should return to immutable upfront design, but which forms of intent must remain explicit while implementation continues iteratively.

B. Generative AI in software engineering: From local generation to systems

Recent work presents Generative AI as a major change in software-engineering practice while emphasizing that its effects vary across tasks, users, and organizational contexts [6]–[8]. Empirical studies of code-generating tools show that programmers work through iterative interaction, verification, and contextual steering rather than simply accepting generated code [9]. This distinction is central to the present study: local code generation may accelerate an activity without resolving cross-file dependencies, architectural boundaries, or team coordination. This makes the relevant bottleneck not code production alone, but the convergence of generated local changes into a shared system design. Software-engineering education is already confronting this change, including questions about how students use and assess Generative AI in project work [10].

C. Upfront planning, architecture, and Spec-Driven Development

Architecture and technical-debt research treats design decisions as evolving constraints whose consequences may accumulate over time. Architectural debt research has developed

concepts and tools for identifying architectural debt, following its accumulation, and relating it to later remediation effort [11], [12]. These traditions motivate measuring early architectural artifacts separately from semantic quality: an observed artifact is evidence of an observable practice, not proof that its design was adequate.

Spec-Driven Development (SDD) extends this concern into AI-assisted workflows. Current practitioner descriptions frame SDD as carrying a specification through planning, task decomposition, implementation, and convergence checks [1], [2]. In this paper, SDD is treated as a testable process proposition: making intent and constraints inspectable may reduce ambiguity when AI tools generate implementation across multiple files and contributors. Existing SDD guidance is used to define the proposition, not as evidence that SDD improves outcomes.

D. Technical debt, rework, and empirical measurement

Technical debt describes the future cost of expedient decisions, but its measurement is difficult because productive iteration can resemble rework in volume-based repository data [13]–[15]. Architecture-debt studies likewise distinguish observable signals from claims about design quality or defect causation. The present work follows that measurement discipline by separating activity volume, clean-path provenance, baseline eligibility, and evaluator outcomes rather than collapsing them into a single score. Similarly, late repository concentration can reflect compensatory recovery, ordinary deadline pressure, delayed integration without quality gain, or project complexity; distinguishing these possibilities requires internal contrasts rather than volume measures alone.

E. Research gap and contribution

The related literature provides strong accounts of Agile coordination, GenAI adoption and interaction, architectural debt, and evolving software practice. What remains less visible is how these perspectives meet in student projects that use Generative AI while being observed across synchronized checkpoints. This study addresses that gap through a bounded mixed-method design combining survey reports, repository activity and path provenance, independent evaluator scorecards, and transcript-based coordination evidence. Its contribution is not a causal estimate of AI productivity or a validation of SDD. It is an evidence-preserving account of how local AI-assisted implementation, temporal integration pressure, explicit architectural artifacts, rework proxies, and independent outcomes can be examined together without silently merging their analytical grains. It also treats ordinary deadline pressure, process heterogeneity, small-sample influence, transcript coverage, and technical complexity as validity threats to be made explicit rather than silently absorbed into the main interpretation.

III. METHODOLOGY

To investigate how planning discipline and coordination evolve in AI-assisted software engineering projects, we designed an empirical study combining longitudinal repository mining, standardized instructor evaluations, multi-checkpoint

student surveys, and qualitative meeting transcripts across two academic offerings.

A. Context, Course Design, and “Rules of the Game”

The study was conducted within a project-based undergraduate software engineering course at a large public university in South America, spanning two semesters: 2025.2 ($N = 9$ analyzable teams) and 2026.1 ($N = 5$ teams), yielding a total inferential dataset of $N = 14$ team-semester (≈ 70 participating students).

Students formed teams of approximately five members through spontaneous self-allocation, with remaining students assigned to vacant slots to ensure balanced team sizes. Teams enjoyed complete autonomy regarding their internal organization, process models, and project management methodologies (e.g., choosing freely between Agile variants, Scrum, Kanban, or unguided approaches). However, course guidelines enforced two structural constraints:

- 1) Teams were required to develop an operational software project featuring an integrated Generative AI component as a core functional requirement, but no particular AI architecture was mandated. Course instruction presented Retrieval-Augmented Generation (RAG)—which grounds Large Language Models (LLMs) in external knowledge bases [16]—as one domain-dependent example rather than a universal requirement. Teams retained autonomy over their architectures, and none of the 14 projects implemented a full RAG pipeline. Instead, they developed project-specific integrations including LLM-supported summarization and content generation, profile or document analysis and matching, conversational tutoring and chatbots, and personalized recommendations, as summarized in Table I.
- 2) Students were strongly encouraged to utilize generative AI assistants throughout all phases of software development, from requirements specification to implementation and testing.

The project lifecycle was anchored by three synchronized temporal checkpoints (T_1 , T_2 , and T_3) paced identically across both cohorts:

- T_1 (**Baseline**): Administered on the first day of class prior to any formal instruction on generative AI processes, capturing baseline perceptions, initial project scope, and early planning commits.
- T_2 (**Mid-Semester**): Conducted following structured instructional modules covering generative AI mechanics, RAG architectures, and software process models, accompanied by the second wave of survey instruments and status presentations.
- T_3 (**Final Integration**): Positioned one week prior to the final project delivery and presentation, capturing the peak pressure of system integration and deadline convergence.

B. Project Scope and Real-World Complexity

To ensure ecological validity and induce realistic coordination friction, the course guidelines explicitly prohibited trivial

algorithmic exercises and toy applications. Under the course rules, teams were required to identify and solve real-world community, educational, or business problems. This constraint forced students to grapple with high domain complexity, authentic user requirements, and the nontrivial challenge of embedding Generative AI into functional architectures.

The resulting software systems spanned diverse domains, ranging from healthcare and financial management to academic research and ecotourism. Table I details the 14 projects analyzed in this study. The breadth and complexity of these applications validate that the observed late-stage integration challenges and chronic coordination friction were driven by authentic software engineering workloads rather than artificially simplistic academic tasks.

C. Data Sources and Collection Framework

Our data collection framework triangulates four distinct evidentiary channels to capture both technical artifacts and human behavioral dynamics.

1. Git Repositories: Complete historical commit logs, branch structures, line change counts, and file-level provenance were extracted from all team repositories across the entire semester lifecycle. Students were explicitly instructed to centralize all project artifacts—including requirements, architectural specifications, documentation, and source code—directly within their Git repositories. This directive defined the repository as the intended source of truth for the team’s production effort. The analysis used read-only repository histories and standardized event tables; Git remains repository evidence, not AI-use telemetry.

2. Independent Evaluator Scorecards: At each checkpoint (T_1 , T_2 , T_3), teams delivered formal status presentations. These were assessed by a panel of independent supervisors (4 evaluators in the 2025.2 cohort, and 3 in the 2026.1 cohort). To prevent groupthink and ensure scoring validity, an independent blind-scoring protocol was enforced: following the presentation and open feedback, each supervisor submitted their scores via a private electronic form. Evaluators remained blind to their peers’ scores throughout the semester. The final project score was calculated as an unweighted average of these independent assessments. Projects were assessed across four structured criteria on explicit bounded scales: *Project Evolution* (0 to 4.0 points), *Scope and Applicability* (0 to 2.0 points), *Technical Complexity* (0 to 2.0 points), and *Engagement and Participation* (0 to 2.0 points). These observations were organized by team-semester and checkpoint for analysis.

3. Multi-Checkpoint Student Surveys: Students completed identical survey instruments at T_1 , T_2 , and T_3 (totaling six instrument cycles). These surveys systematically captured data across five core constructs: Demographics, Professional Experience, GenAI Adoption, Perceptions of AI Impact on SE Roles, and Project Expectations. To ensure replicability, the core dimensions and sample variables tracked across these checkpoints are detailed in Table II. Survey responses are cohort-by-semester/checkpoint data and carry no team

TABLE I
ANALYZED STUDENT PROJECTS BY COHORT, DEMONSTRATING REAL-WORLD PROBLEM SCOPE AND GENERATIVE AI INTEGRATION

Semester	Project Name	System Description & AI Integration Scope
2026.1	FoodGuard	Web platform utilizing nutritional data and medical history to identify food allergens and prevent adverse reactions.
2026.1	RevisAI	Educational system that generates intelligent flashcards and organizes customized weekly study review cycles.
2026.1	SmartFlowIA	SaaS WhatsApp chatbot focusing on intelligent onboarding, orientation, and welcoming for university freshmen.
2026.1	Interview Prep IA	AI-driven system that analyzes resumes against job vacancies, simulates interviews, and provides personalized candidate feedback.
2026.1	EnemBot	Domain-specific chatbot designed to tutor and assist students with questions regarding the Brazilian National High School Exam.
2025.2	CampusGuide	Intelligent virtual assistant and campus navigation system designed to help students and professors locate university resources.
2025.2	EstudaAI	Study management platform that processes multimodal files (audio, image, PDF) to generate lecture summaries, reminders, and track grades.
2025.2	CareerPath IA	Resume analysis tool that uses GenAI to match user profiles with market demands and suggest targeted skill development.
2025.2	Buliçoso	Healthcare application that utilizes LLMs to simplify complex medical leaflets and manage medication reminders for the elderly.
2025.2	EcoRoteiro	Ecotourism platform that generates personalized travel itineraries based on user profiles, physical conditions, and specific interests.
2025.2	Research Flow	Academic research assistant that searches articles, generates intelligent summaries, and formats texts to academic standards (e.g., ABNT, APA).
2025.2	ExamForge	Assessment simulator that ingests past exams and course materials to generate new tests matching specific professor grading patterns.
2025.2	SoFiA	Financial management web platform utilizing AI to help university students dynamically organize their personal finances.
2025.2	Synapse	Requirements engineering tool that summarizes technical client meetings to automatically generate functional requirements, user stories, and backlogs.

identifier; Self-Reported AI Usage and Perceptions (M1) and Perceived Role Disruption (M2) therefore remain at that grain.

4. Qualitative Meeting Transcripts: To contextualize coordination dynamics and evaluative feedback, we constructed a bounded, single-cohort qualitative corpus from audio recordings of the three formal status presentation checkpoints in 2025.2. The corpus comprises the full audio records of the student presentations alongside the subsequent deliberative discussions among the supervisors in a private chat group. Crucially, these supervisory discussions were free-form qualitative reflections, completely decoupled from the numeric scorecasting process. To make the transcription data computationally feasible for analysis, the raw text was segmented into analytical chunks, yielding 100 distinct fragments (35 at T_1 , 37 at T_2 , and 28 at T_3). Coordination-Language Evidence (M5) reconstructs one global corpus per checkpoint; chunks and source sessions are provenance units, not independent observations. This qualitative deep dive informs interpretation of the repository evidence collected across both cohorts, but it is not used to claim cross-cohort replication: no transcript-based comparison with 2026.1 is made, and no transcript evidence is imputed for that cohort.

Data Anonymization and Privacy: To ensure strict ethical compliance and protect participant confidentiality, all raw data streams were subjected to a rigorous anonymization pipeline, combining deterministic heuristics with LLM-assisted redaction to scrub unstructured text across commit messages, survey free-text responses, and audio transcripts. Finalized analytical artifacts exclude direct personal identifiers. Detailed call

records and raw payload provenance remain restricted to the internal reproducibility record.

D. Statistical and Analytical Concepts

The analysis uses descriptive summaries to preserve the scale and grain of each evidence stream. Means are reported with standard deviations when the metric is an approximately interval-like score; medians and interquartile ranges are used for skewed counts and ratios. Proportions retain their denominators, and rolling seven-day windows are indexed relative to evaluator anchors; overlapping windows are therefore summaries of nearby periods, not independent observations.

For exploratory associations, Spearman’s rank correlation ρ measures the strength and direction of a monotonic relationship between two ranked variables without assuming a linear scale [17]. Its handling of ties is retained in the metric definition; study-specific limits from sample size, missingness, and influential observations are addressed in Threats to Validity. Artifact–Rework–Outcome Associations (M9) reports exploratory p-values as descriptive diagnostics rather than confirmatory significance tests. Leave-one-out analysis recomputes an association after removing each eligible team-semester in turn; the resulting range shows sensitivity to influential cases and is not a confidence interval. These summaries are always interpreted with their denominator, stratum, and measurement-availability state.

E. Advanced Metrics Operationalization

Sections IIIA–D established what was collected. This subsection formalizes how repository history, evaluator score-

TABLE II
STRUCTURE OF THE MULTI-CHECKPOINT SURVEY INSTRUMENT

Construct	Evaluated Variables & Inquiries	Data Type
Demographics & Academic Background	Age, Gender, Current academic semester, Previous degrees, High school educational background.	Categorical
Professional SE Experience	Years of software development experience, Current employment status, Preferred engineering role (e.g., Backend, QA, ML), Largest past team size.	Categorical
Generative AI Adoption & Tooling	Frequency of AI usage, Specific tools adopted (e.g., ChatGPT, Gemini, GitHub Copilot, Claude), Primary use cases (e.g., Code generation, Debugging, Documentation).	Categorical & Multiple Choice
Perceived AI Impact on SE Roles	Perceived risk of extinction or severe disruption for specific roles: Backend, Frontend, QA, Project Manager, Product Manager, Scrum Master.	5-point Likert
Expectations & Motivation	Confidence levels prior to project initiation, Motivation to learn new techniques, Expected 5-year impact of AI on the industry.	5-point Likert & Open-ended

cards, survey responses, and transcript text become nine metric families used for RQ1–RQ3. Each family retains its analytical grain and exposes components only when they represent distinct constructs; no component is silently combined into a new score. To improve readability, we retain the M1–M9 identifiers for traceability but refer to each family by a concise descriptive name throughout the results and discussion (for example, Self-Reported AI Usage and Perceptions (M1) and Repository Activity (M3)).

A note on $T_1/T_2/T_3$ for repository-derived measures. The checkpoint labels for commit and file history denote windows bounded by each team’s evaluator presentation dates, not single calendar days. This distinction is retained throughout the repository-derived families and is made explicit in the corresponding artifacts and provenance comments.

In addition to the primary M1–M9 metric families, we compute robustness and sensitivity views that preserve the same source data and analytical grains: non-overlapping phase bins for repository activity and clean churn, final-seven-day concentration contrasts by evaluator-score change and planning scope, operational-regularity proxies, technical-complexity profiles, and leave-one-out influence diagnostics. These views qualify interpretation and validity threats; they do not introduce new causal estimators.

1) *RQ1: How do student developers utilize Generative AI, and how does this adoption shape their longitudinal perceptions of software engineering roles and project expectations?*: RQ1 is operationalized at cohort-by-semester/checkpoint grain because the anonymized survey responses contain no team identifier. Self-Reported AI Usage and Perceptions (M1) separates the perception panel from self-reported usage frequency, tasks, tools, prior experience, autonomy/tool balance, and exploratory career topics. Perceived Role Disruption (M2) preserves role-specific Likert distributions and paired sensitivity summaries. These families describe self-report and perceived role disruption; neither is AI-use telemetry or objective extinction risk.

Self-Reported AI Usage and Perceptions (M1). The six perception families remain separate. Descriptive distributions are reported alongside raw usage fields and missingness; no heterogeneous family average is interpreted as a behavioral

dependency measure. Career-topic coding remains exploratory.

Perceived Role Disruption (M2). Role-specific Likert responses are summarized by cohort, checkpoint, and role. Where privacy-preserving respondent identifiers permit it, paired T1–T3 sensitivity summaries are reported without exposing identities.

2) *RQ2: How does the temporal density of repository activity contrast with the qualitative typification of human coordination friction across the project lifecycle?*: **Repository Activity (M3).** This family separates phase activity, author concentration, and rolling seven-day activity. A secondary pooled relative-window view removes the semester dimension only for visualization; it does not replace semester-stratified observations. A mutually exclusive phase-bin view and final-seven-day phase shares provide a robustness check against interpreting overlapping rolling windows as independent observations.

Clean-Change Dynamics (M4). This family separates clean-change magnitude, touching-commit intensity, artifact composition, and rolling clean churn. The consistent path-classification rule defines the included source and test paths; commit counts remain contextual rather than the primary Clean-Change Dynamics (M4) outcome. Clean source/test churn is also summarized in the same non-overlapping phase structure used for Repository Activity (M3), allowing the late-stage concentration pattern to be inspected under a clean-path policy.

Coordination-Language Evidence (M5). This family reconstructs one global transcript corpus per checkpoint and deterministically counts versioned lexical markers for alignment, handoff, integration, blockers, and rework. It reports density, composition, corpus coverage, and an auditable chunk queue. Coordination-Language Evidence (M5) is not joined or correlated with team-semester Repository Activity (M3) or Clean-Change Dynamics (M4) observations. Transcript-derived M5 evidence is available only for 2025.2; 2026.1 is treated as unavailable coverage, not as zero marker evidence.

3) *RQ3: How does the observable scope and content evidence of upfront architectural artifacts at baseline T_1 associate with late-stage clean-path rework and independent, blind-scored project outcomes?*: **Upfront Architec-**

ture Evidence (M6). Architectural Artifact Scope (M6a) reports T1 artifact presence and repository-visible scope. Approved Planning Content Evidence (M6b) reports separate categories—goals, architecture/design, task decomposition, risk/dependency, and literal evidence quotes—for nine observed team-semesters; five cases without T1 commit subjects were not measured. Planning Content Evidence (M6b) is exploratory and is never reduced to a composite planning-quality score.

Repository Inactivity (M7). This family reports recent seven-day repository inactivity around evaluator checkpoints and in T3-relative rolling windows. It describes repository timing and coverage only; it does not identify planning omission and is not an independent predictor in Artifact–Rework–Outcome Associations (M9). Operational-regularity summaries extend this timing view by combining observable temporal and author-activity dispersion in Git. They are descriptive process proxies, not validated measures of agile adherence, productivity, or process quality.

Clean-Path Rework (M8). This family separates clean rework magnitude, the clean rework ratio, the late-stage rolling trajectory, and baseline eligibility. This is a path-provenance proxy for destructive rework, not semantic defect validation. A path first observed at T1 or T2 is eligible for the provenance analysis; baseline-ineligible cases remain separate and are not imputed as stability or low quality.

M9: associations among artifacts, rework, and outcomes. This family reports descriptive Spearman associations between T1 architectural-artifact predictors and Clean-Path Rework (M8a/M8b) or independent, blind-scored T3 evaluator outcomes, restricting ratio analyses to the baseline-eligible stratum. Approved Planning Content Evidence (M6b) fields enter a separate non-composite association output for observed cases. Leave-one-out diagnostics expose influence. Technical-complexity profiles use evaluator technical-complexity scores and repository-structural heuristics as contextual variables for sensitivity summaries. They are not causal adjustments, but they help assess whether rework and concentration patterns may be partly explained by project difficulty. No Repository Inactivity (M7) predictor or floor-imputation analysis is used.

Table III summarizes the nine metric families, their components, and their primary analytical grains.

F. LLM Usage in Data Extraction, Processing, and Metrics

Language-model assistance is used selectively for data extraction and is kept separate from deterministic metric construction. Self-Reported AI Usage and Perceptions (M1) uses coded survey responses, and Coordination-Language Evidence (M5) uses deterministic lexical coding of the transcript corpus; neither metric receives a new language-model score. Planning Content Evidence (M6b) uses structured extraction from T_1 commit subjects for nine observed team-semesters; five team-semesters have no T_1 subjects and remain unavailable rather than being imputed.

Table IV summarizes the metric-bound language-model task and the metric component it supports.

TABLE III
SUMMARY OF METRIC FAMILIES AND COMPONENTS

ID	Metric family and components	Primary grain
M1	Self-Reported AI Usage and Perceptions	Cohort $\times$ checkpoint
M2	Perceived Role Disruption	Cohort $\times$ checkpoint $\times$ role
M3	Repository Activity	Team-semester / window
M4	Clean-Change Dynamics	Team-semester / checkpoint / window
M5	Coordination-Language Evidence	Global corpus / checkpoint
M6	Upfront Architecture Evidence	Team-semester
M7	Repository Inactivity	Team-semester / window
M8	Clean-Path Rework	Team-semester / window
M9	Artifact–Rework–Outcome Associations	Team-semester / stratum

TABLE IV
METRIC-BOUND LANGUAGE-MODEL TASK

Task	Feeds	Structured output
Planning evidence extraction	Planning Content Evidence (M6b)	Goals, architecture/design, task decomposition, risk/dependency, and literal evidence quotes from T_1 commit subjects.

Planning Content Evidence (M6b) outputs were reviewed by the research team and retained for exploratory analysis as separate categories. They are not reduced to a composite planning-quality or architectural-quality score.

Persistence and auditability. Model-assisted outputs are retained with their associated prompts, responses, hashes, usage metadata, and status so that each reported value can be traced to its source evidence. These records support reproducibility while remaining outside the paper-facing narrative.

Inference boundary. All reported metric families and components are descriptive or exploratory. Missing evidence remains distinct from zero or low quality, incompatible analytical grains are not silently combined, and no causal claim is made without a design that supports it.

IV. RESULTS

This section reports the descriptive and exploratory results produced by the validated metric artifacts. Results are organized by research question and retain the analytical grain of each metric; no claim combines incompatible grains or treats self-reported evidence as behavioral telemetry.

A. RQ1: How do student developers utilize Generative AI, and how does this adoption shape their longitudinal perceptions of software engineering roles and project expectations?

RQ1 is supported by two distinct evidence streams: Self-Reported AI Usage and Perceptions (M1), which summarizes six perception families and separately reports self-reported AI use, and Perceived Role Disruption (M2), which summarizes perceived disruption agreement for six software-engineering roles. All six cohort-by-semester/checkpoint cells in the M1 family have complete item coverage, with 46, 41, and 40 respondents in 2025.2 and 22, 20, and 18 respondents in 2026.1 at T_1 , T_2 , and T_3 , respectively. These are repeated cohort summaries, not an identified individual longitudinal panel.

The six perception families do not move as a single trajectory. These item means use the inherited 0–4 score scale, except for the separately reported autonomy/tool-balance item, which uses its observed 1–5 response scale. Values below are reported as mean $\pm$ SD . In 2025.2, mean AI benefit increases from 2.13 ± 1.00 at T_1 to 2.25 ± 0.90 at T_3 , while five-year career impact decreases from 1.48 ± 1.01 to 1.33 ± 0.83 . Autonomy/tool balance changes from 3.09 ± 0.89 to 3.23 ± 0.95 . In 2026.1, AI benefit decreases from 2.00 ± 1.07 to 1.56 ± 0.98 , whereas autonomy/tool balance increases from 2.64 ± 0.85 to 3.44 ± 0.92 , and five-year career impact changes from 1.55 ± 0.91 to 1.67 ± 0.91 . The project-expectation families remain comparatively low: at T_3 , project challenges average 0.15 ± 0.36 in 2025.2 and 0.06 ± 0.24 in 2026.1, while project feeling averages 0.35 ± 0.77 and 0.50 ± 0.86 , respectively. These values describe distinct response families and should not be interpreted as one latent measure of AI dependence.

Figure 1 presents these six families with cohort-specific standard-deviation bars. Table V reports the corresponding endpoint values in compact form.

The raw usage fields provide a separate adoption description. The share reporting daily general AI use rises from 41.3% to 45.0% in 2025.2 and from 54.5% to 61.1% in 2026.1 between T_1 and T_3 . For software projects specifically, the daily-use share changes from 23.9% to 32.5% in 2025.2 and from 40.9% to 38.9% in 2026.1. The most frequently reported tasks include general research/questions, code generation, debugging, and documentation. For example, code generation rises from 65.2% to 77.5% in 2025.2, while documentation is reported by 72.7% of the 2026.1 T_1 respondents and by 66.7% at T_3 . These are self-reported frequencies and selected activities, not instrumented measures of tool use.

Reported tool use is concentrated in a small group of widely named tools but is not constant across cohorts. ChatGPT is reported by all 2025.2 respondents at T_1 and T_3 , and by 83.3% of the 2026.1 T_3 respondents. Gemini is reported by 77.5% of 2025.2 respondents at T_3 and 77.8% of 2026.1 respondents at T_3 . Prior AI-project experience also increases across the observed checkpoints, from 41.3% to 70.0% in 2025.2 and from 31.8% to 94.4% in 2026.1. Because the respondent

counts and identities are not stable individual panels, these changes cannot be interpreted as within-person learning or adoption effects.

Career-impact topic coding is retained as an exploratory supplement rather than a validated result. Its largest categories are uncertainty or conditional statements and other/insufficient evidence; for example, together they account for 56.9% of topic assignments at 2025.2/ T_1 and 65.2% at 2026.1/ T_3 . The coding is rule-based and explicitly unvalidated, so these proportions are presented as exploratory descriptions only.

Perceived Role Disruption (M2) shows a role-specific pattern rather than a uniform increase in perceived disruption. The values below are means on the 1–5 Likert agreement scale; each parenthetical value is the corresponding SD . In 2025.2, Frontend has the highest mean agreement at all three checkpoints: 2.74 ± 1.10 , 2.68 ± 1.08 , and 2.73 ± 1.18 . Project Manager remains among the lowest: 1.83 ± 1.02 , 1.83 ± 0.86 , and 2.08 ± 0.89 . In 2026.1, Frontend is also high, moving from 3.14 ± 1.13 to 2.65 ± 1.09 and 3.28 ± 1.36 ; Scrum Master starts at 3.14 ± 1.25 at T_1 . Agreement is therefore role-specific and cohort-dependent; it does not support a general claim that all roles become progressively more exposed to replacement or transformation. Figure 2 provides the complete role-by-cohort/checkpoint view; its color scale is fixed to the 1–5 response range.

The privacy-preserving paired sensitivity analysis reinforces this caution. There are 38 paired respondents per role in 2025.2 and 16 per role in 2026.1. Changes are mixed rather than uniformly directional: in 2025.2, Project Manager shows the largest increase (+0.32) and Backend is nearly unchanged (−0.03); in 2026.1, QA increases by +0.69, while Scrum Master decreases by −0.31. These paired values are sensitivity summaries.

B. RQ2: How does the temporal density of repository activity contrast with the qualitative typification of human coordination friction across the project lifecycle?

RQ2 contrasts three descriptive evidence streams that operate at different grains. Repository Activity (M3) measures activity timing, author participation, and concentration for 14 team-semesters. Clean-Change Dynamics (M4) measures clean source/test change at team-semester and checkpoint grain. Coordination-Language Evidence (M5) counts candidate coordination markers within a global transcript corpus by checkpoint. The measures are therefore reported side by side as complementary descriptions, not as a common-scale effect or a statistical correlation.

Repository Activity (M3) shows a pronounced concentration around the final evaluator anchor. In the semester-pooled rolling view, total commits increase from 95 at 21 days before the anchor to 159 one day before, peak at 302 one day after, and fall to 28 at seven days after. At the anchor itself, all 14 team-semesters have observed activity; at −7 days, only 5 of 14 are active, and at +7 days, 7 of 14 remain active. The corresponding median commit count per team-semester is 12.5 at the anchor and 0.5 at +7 days. These rolling windows

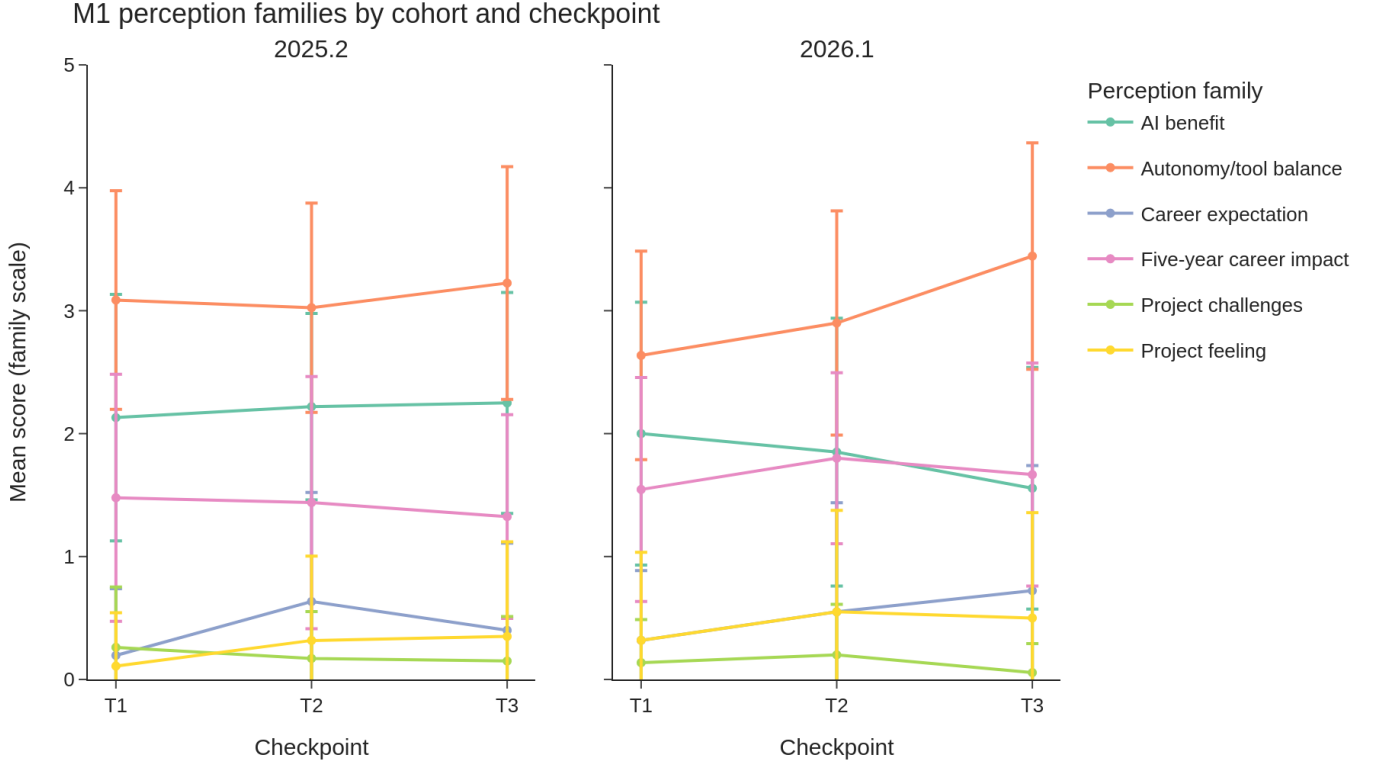


Fig. 1. Self-Reported AI Usage and Perceptions (M1) by cohort and checkpoint. Points show means and bars show one standard deviation; family scales are retained from the underlying instruments.

TABLE V
SELF-REPORTED AI USAGE AND PERCEPTIONS (M1): ENDPOINT MEANS $\pm$ SD

Family	2025.2 T_1	2025.2 T_3	2026.1 T_1	2026.1 T_3
AI benefit	2.13 $\pm$ 1.00	2.25 $\pm$ 0.90	2.00 $\pm$ 1.07	1.56 $\pm$ 0.98
Five-year career impact	1.48 $\pm$ 1.01	1.33 $\pm$ 0.83	1.55 $\pm$ 0.91	1.67 $\pm$ 0.91
Autonomy/tool balance	3.09 $\pm$ 0.89	3.23 $\pm$ 0.95	2.64 $\pm$ 0.85	3.44 $\pm$ 0.92
Project challenges	0.26 $\pm$ 0.49	0.15 $\pm$ 0.36	0.14 $\pm$ 0.35	0.06 $\pm$ 0.24
Project feeling	0.11 $\pm$ 0.43	0.35 $\pm$ 0.77	0.32 $\pm$ 0.72	0.50 $\pm$ 0.86

overlap, so adjacent points are temporal summaries rather than independent observations. Figure 3 visualizes this pooled temporal shape; the dashed line marks the final evaluator anchor.

The checkpoint participation view shows the same concentration with substantial heterogeneity. Across the 14 team-semesters, 7 have any post-checkpoint activity. The final-window share of all commits ranges from 9.5% to 100.0%; three team-semesters have fewer than ten total commits, and their shares should be treated as small-denominator descriptions. In the author concentration output, the median pre-anchor window contains 12.5 commits, 3.5 active days, and 3 authors, with a median maximum-author share of 0.50 and median Gini coefficient of 0.22. Post-anchor windows are frequently inactive; where activity exists, concentration statistics are not interpreted for no-activity cases.

The same final-stage pattern remains visible when the 91 days before T_3 are re-expressed as mutually exclusive

phases rather than overlapping rolling windows. In the score-trajectory stratification, the eight improved team-semesters assign a mean of 56.7% of commits and 59.0% of clean source/test churn to the final seven days, while the four stable team-semesters assign 24.6% and 18.2%, respectively. The two declined team-semesters are too few for a stable contrast, but their final-seven-day shares are 32.9% for commits and 45.0% for clean churn. These phase-share summaries are descriptive checks against the overlap structure of the rolling M3 view, not independent causal tests. Figures 4 and 5 show the commit and clean-churn variants.

To address whether the final peak is simply “student syndrome,” we also compare final-seven-day concentration with evaluator-score change and repository-visible planning scope at team-semester grain ($n = 14$). The median final-seven-day concentration cut is 43.7% for commits and 35.0% for clean churn; the descriptive score-change threshold is 0.125 on the composite evaluator-score scale. Higher final

M2 perceived role disruption agreement

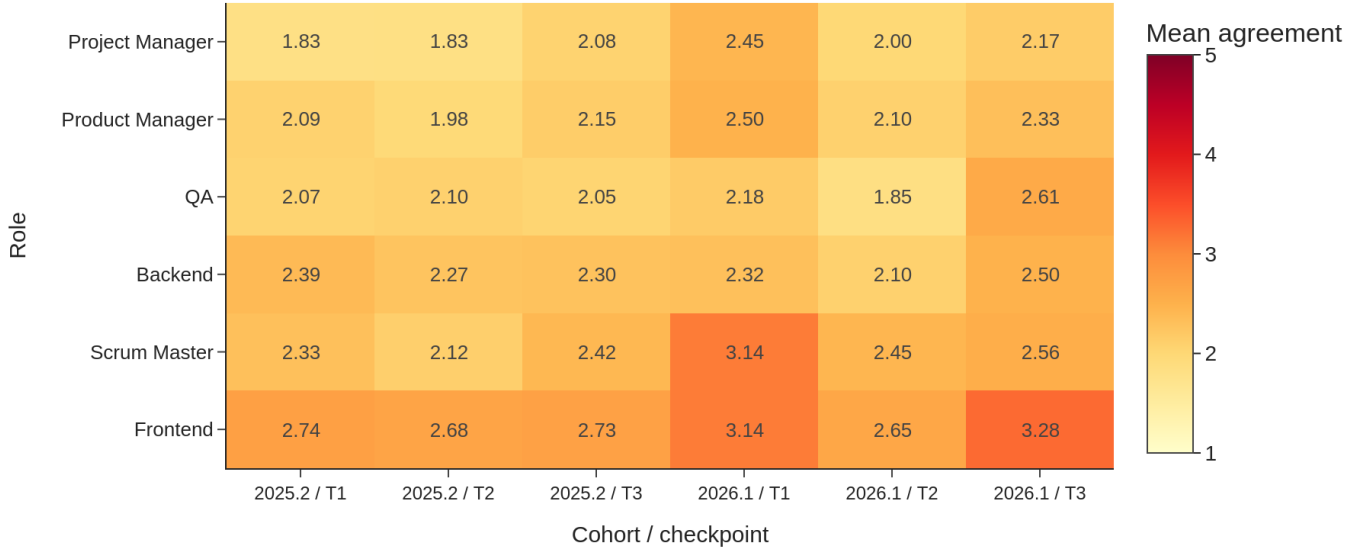


Fig. 2. Perceived Role Disruption (M2) by cohort and checkpoint. Cells show mean agreement on the 1–5 Likert scale.

concentration appears in both positive-delta and non-positive-delta cases. For example, among commit concentration quadrants, five team-semesters combine higher final concentration with positive score change, while two combine higher final concentration with no score gain. Planning-scope quadrants similarly separate late concentration from outcomes: lower-planning/higher-final-concentration cases have lower mean T3 score than lower-planning/lower-concentration cases in the commit view (2.16 versus 2.45), but the same summaries are small-n descriptions and not an estimator of planning quality or procrastination. Figures 6–9 make these contrasts explicit for commits and clean churn.

Operational regularity provides a final repository-visible process check. The regularity index is descriptive, with larger values indicating activity more distributed across the observed Git timeline; it is not a validated agile process scale. Figure 10 shows that both low and high final concentration appear across the observed regularity range, so the late-activity interpretation should remain bounded to Git evidence rather than treated as a direct measure of team process maturity.

Clean-Change Dynamics (M4) reproduces the temporal increase using a narrower clean-change definition. Pooled clean churn rises from 21,718 lines across 7 observed team-semesters at T_1 , to 47,545 lines across 10 at T_2 , and 136,325 lines across all 14 at T_3 . The median clean churn per team-semester rises from 66.5 lines at T_1 to 3,232.5 at T_2 and 6,975.5 at T_3 . Clean change accounts for 42.9%, 19.6%, and 1.0% of all recorded churn at the three checkpoints, respectively, because excluded or non-measurement paths dominate the raw line-change total in the final checkpoint. This distinction is why Clean-Change Dynamics (M4) is not interchangeable with total repository activity.

Coordination-Language Evidence (M5) provides a separate textual axis for the 2025.2 transcript corpus. The corpus contains 35, 37, and 28 transcript chunks at T_1 , T_2 , and T_3 , respectively, with two source sessions at each checkpoint. Candidate marker density increases from 0.515 to 1.132 and 1.287 markers per 1,000 transcript tokens. Marker counts are 12, 54, and 40, over 23,296, 47,705, and 31,087 tokens. The composition also changes: integration markers account for 20 of 54 markers at T_2 , while blockers account for 7 of 40 at T_3 . These are deterministic lexical matches in a global corpus, not diagnoses of coordination failure; transcript coverage is unavailable for 2026.1.

Table VI consolidates the principal RQ2 values while retaining their distinct units and analytical grains. Figure 12 shows the Clean-Change Dynamics (M4) and Coordination-Language Evidence (M5) checkpoint trajectories in separate panels.

Because M5 is available only for 2025.2, Figure 11 reports the triangulation as a coverage-aware 2025.2 panel rather than zero-filling the unavailable 2026.1 semester. Within the nine 2025.2 team-semesters, median clean churn rises from zero at T_1 to 2,803 at T_2 and 5,837 at T_3 , while the median composite evaluator score rises from 2.00 to 2.22 and 2.41. The M5 density line remains global to the transcript corpus, so this panel is a contextual triangulation rather than a team-level M5 association.

Taken together, Repository Activity (M3) and Clean-Change Dynamics (M4) show that repository-visible activity and clean source/test change become most concentrated near the final checkpoint, while Coordination-Language Evidence (M5) shows an increase in candidate coordination-language density across the observed 2025.2 transcript corpus. The three pat-

M3 pooled rolling repository activity

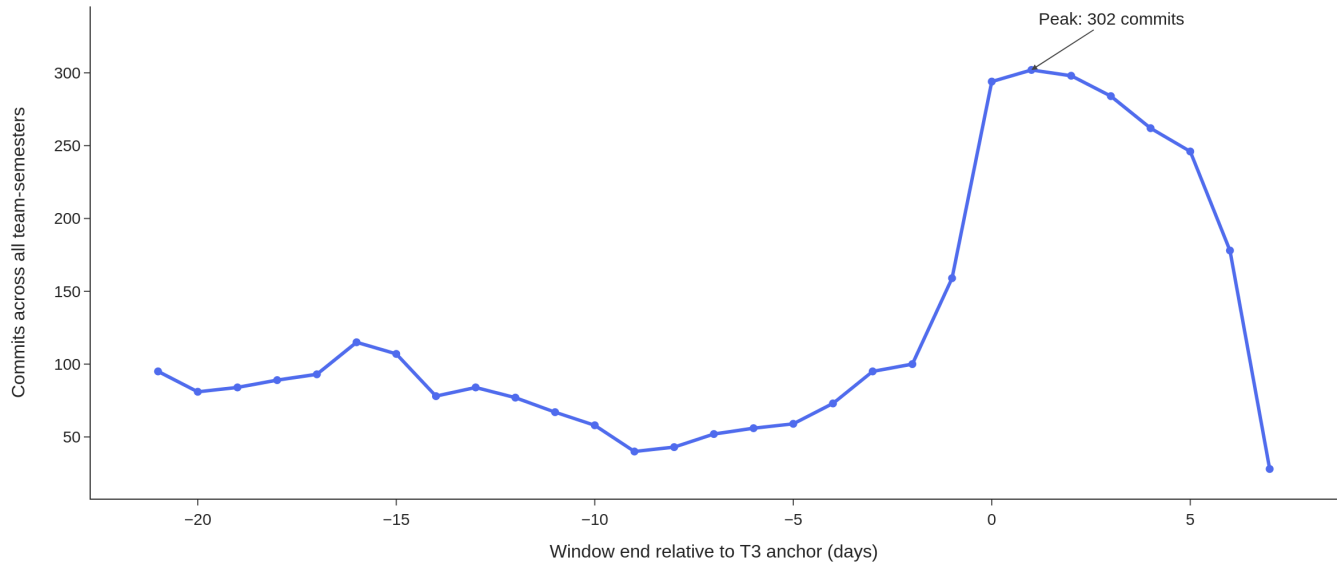


Fig. 3. Repository Activity (M3) in pooled rolling windows. The vertical marker identifies the T_3 evaluator anchor; the annotation identifies the pooled peak.

TABLE VI
RQ2 TEMPORAL SUMMARY WITH EXPLICIT UNITS

Metric	Unit / grain	T_1 or -7	T_2 or -1	T_3 or 0	Post-anchor	Coverage
Repository Activity (M3)	Pooled commits / rolling window	52 (-7)	159 (-1)	294 (0)	302 (+1); 28 (+7)	14 team-semesters
Clean-Change Dynamics (M4)	Clean churn lines / checkpoint	21,718	47,545	136,325	–	7 / 10 / 14 observed
Coordination-Language Evidence (M5)	Markers per 1,000 tokens / global corpus	0.515	1.132	1.287	–	2025.2 only

terns are compatible with the RQ2 descriptive contrast within the observed evidence streams.

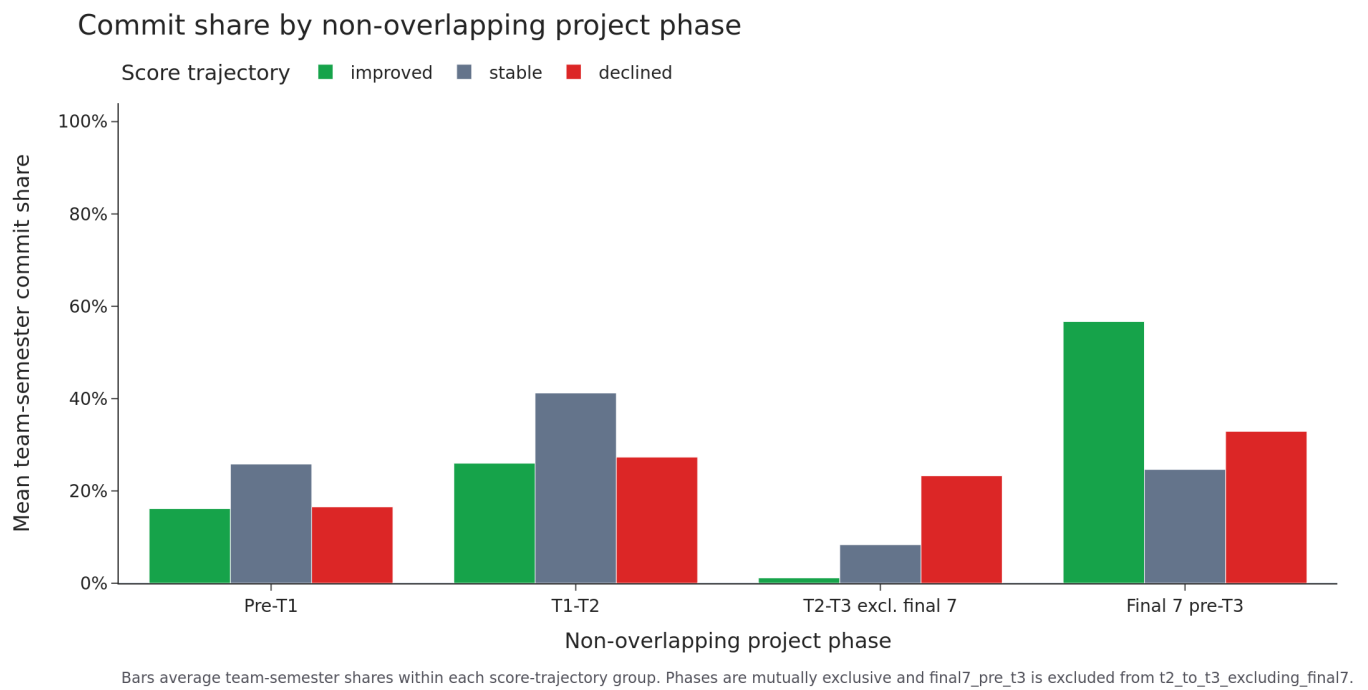


Fig. 4. RQ2 non-overlapping phase shares for commits by evaluator-score trajectory. Each team-semester contributes phase shares that sum to 100%; the final phase is the seven-day pre- T_3 window.

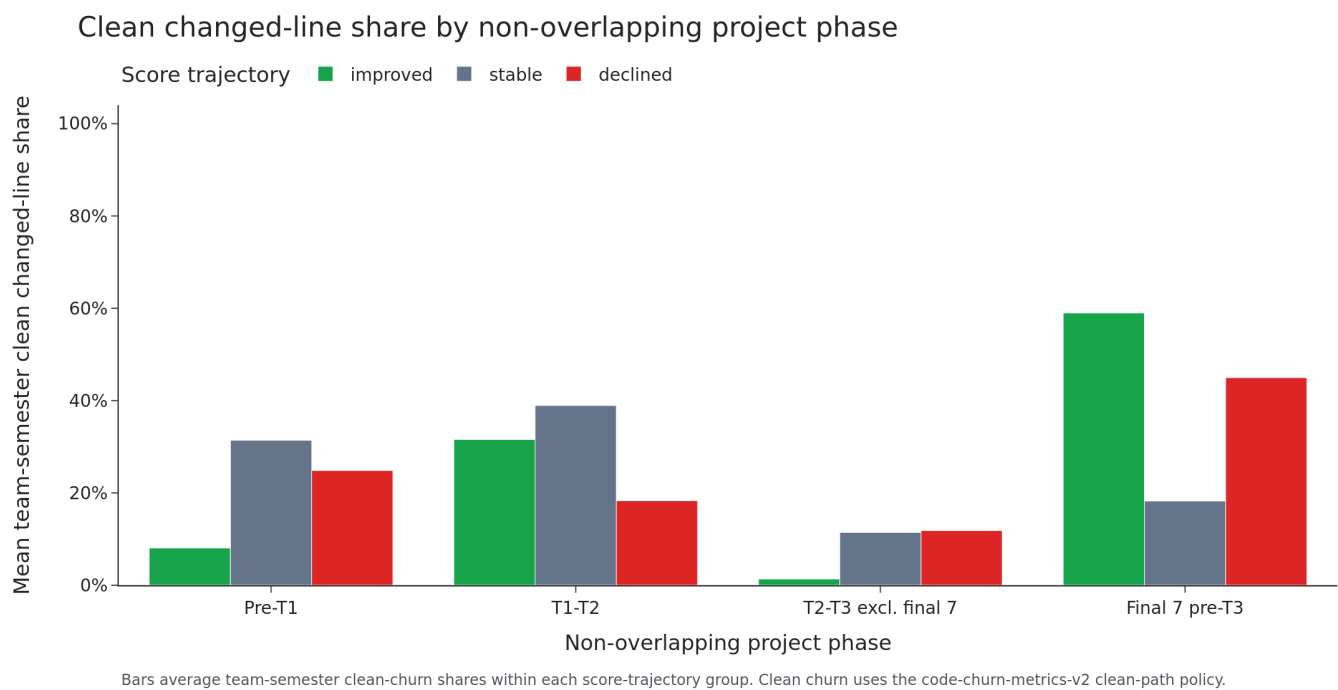


Fig. 5. RQ2 non-overlapping phase shares for clean source/test churn by evaluator-score trajectory. The panel tests whether late concentration remains visible under mutually exclusive phases and the clean-path policy.

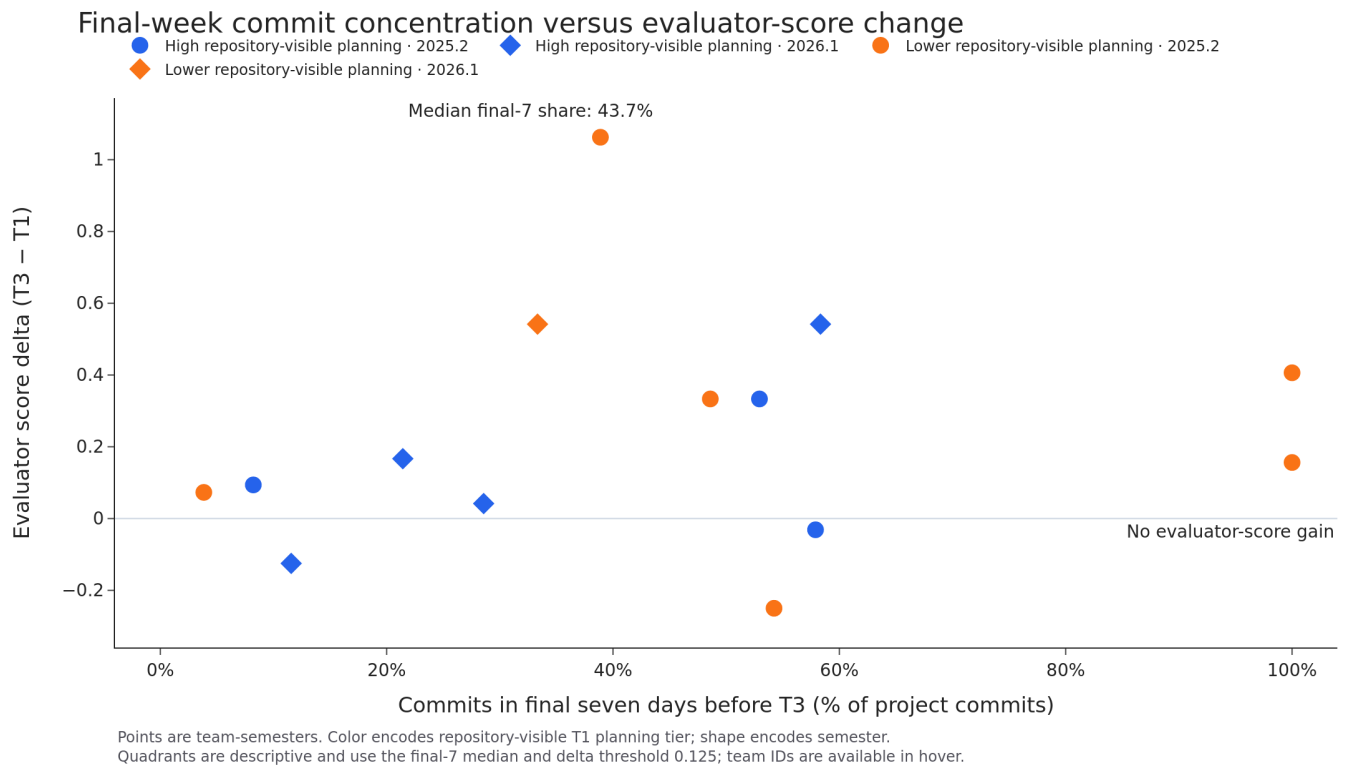


Fig. 6. Evaluator-score change versus final-seven-day commit concentration ($n = 14$ team-semester). Color encodes repository-visible planning tier; the figure separates late activity with score gain from late activity without score gain.

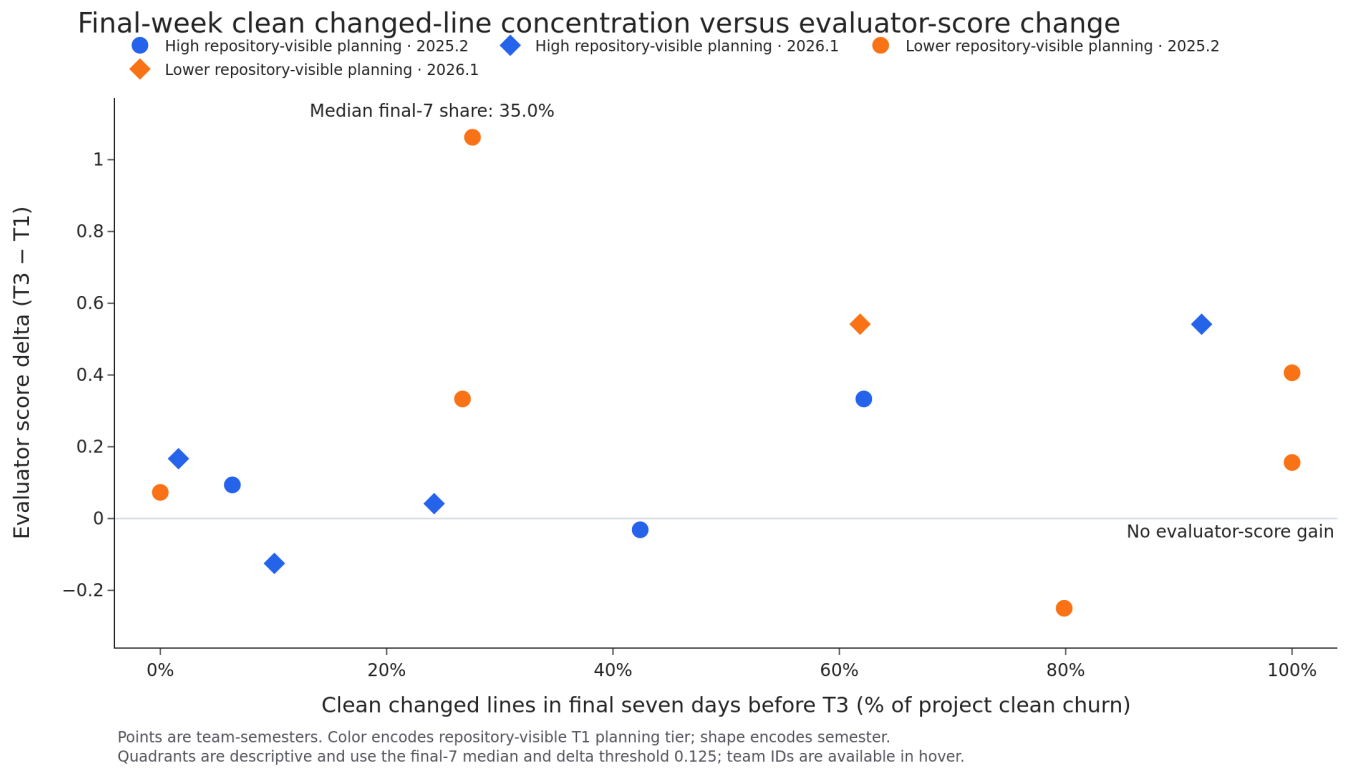


Fig. 7. Evaluator-score change versus final-seven-day clean source/test churn concentration ($n = 14$ team-semesters). Clean churn narrows the activity view to source/test paths retained by the clean-path policy.

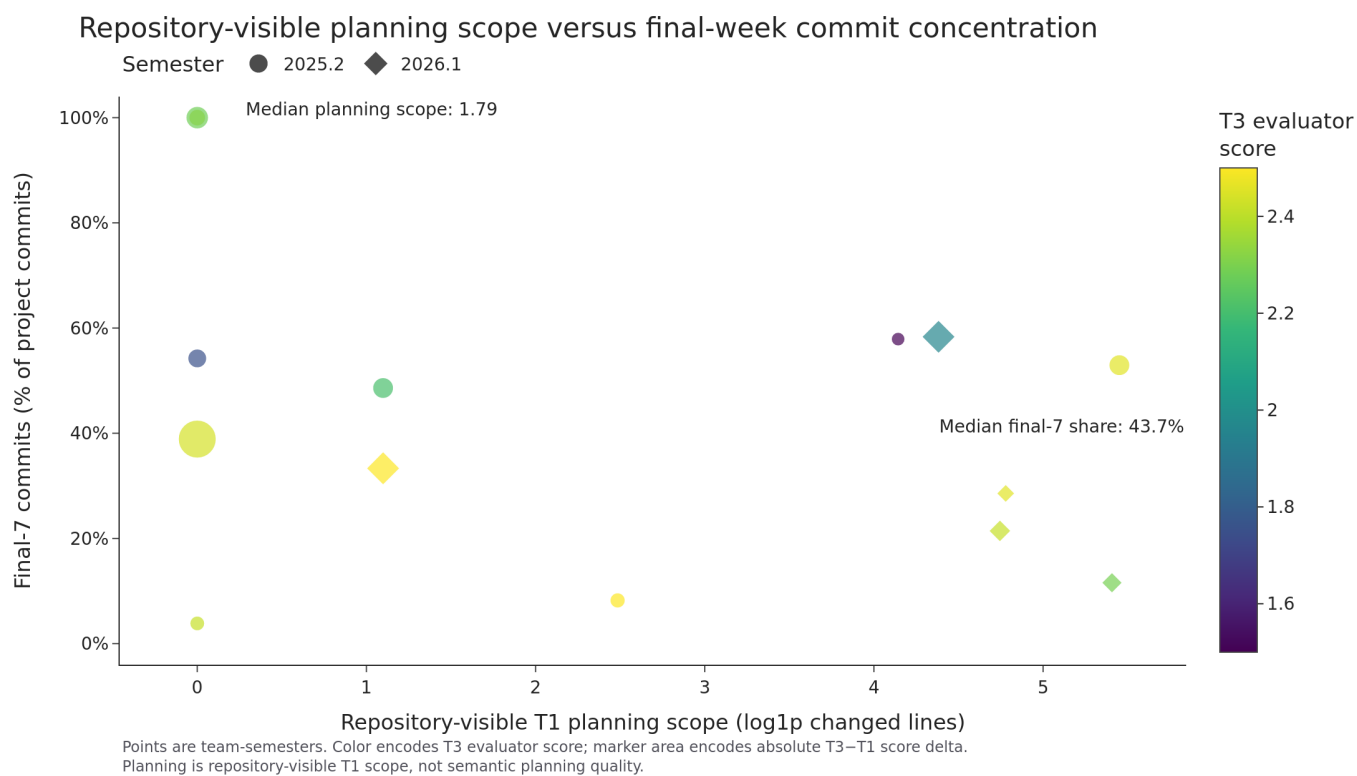


Fig. 8. Repository-visible planning scope versus final-seven-day commit concentration. Marker color reports T3 evaluator score and marker size reports absolute score change; quadrants are descriptive median splits.

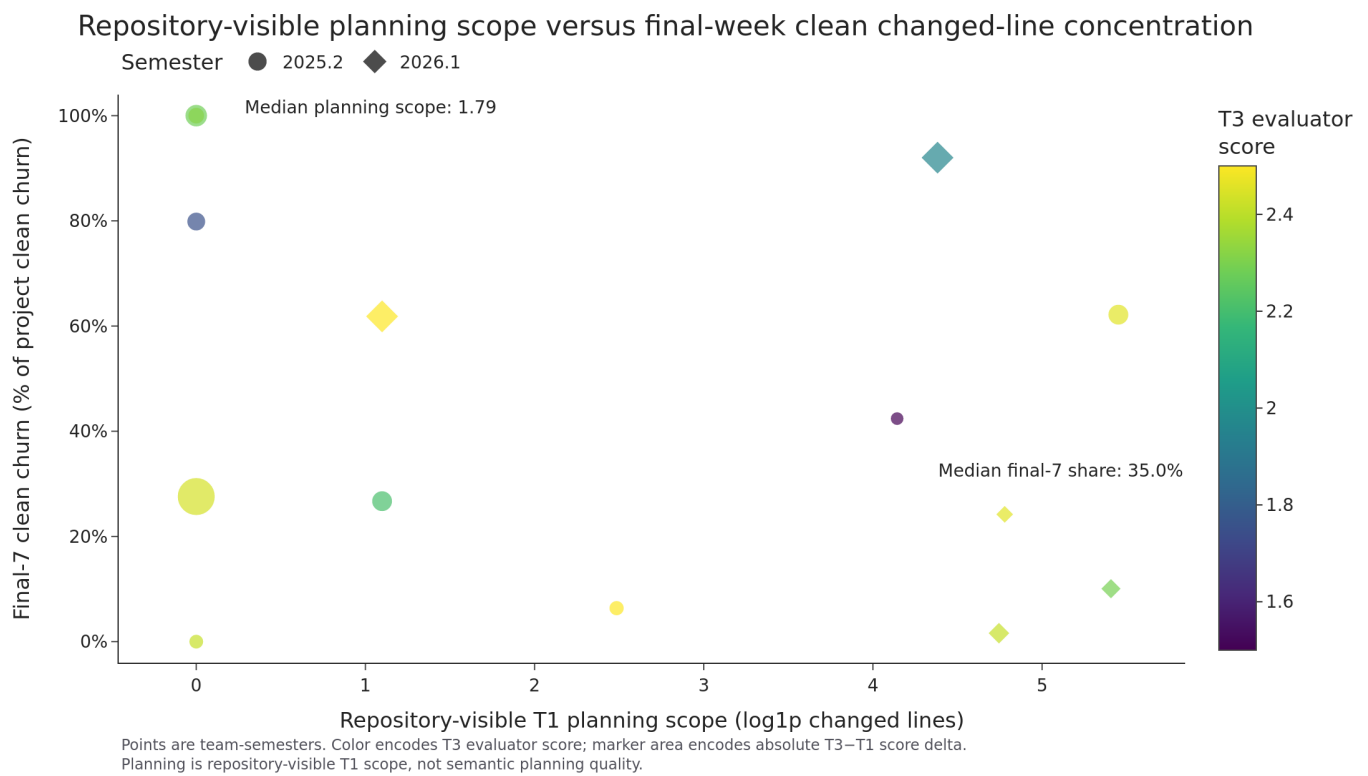


Fig. 9. Repository-visible planning scope versus final-seven-day clean source/test churn concentration. Planning scope is structural and repository-visible, not a semantic planning-quality score.

Operational regularity versus final-week concentration

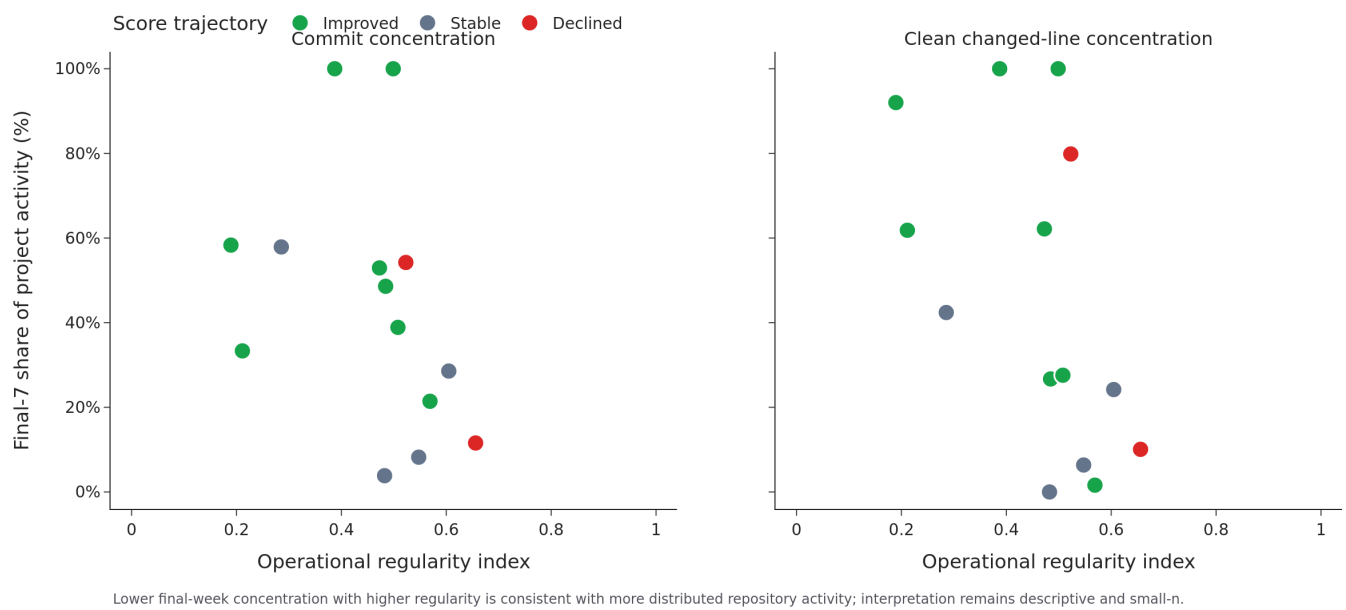


Fig. 10. Operational regularity versus final-seven-day concentration. The index summarizes repository-visible temporal and author regularity and should not be read as a validated agile-process or productivity metric.

Coverage-aware M5 triangulation for 2025.2

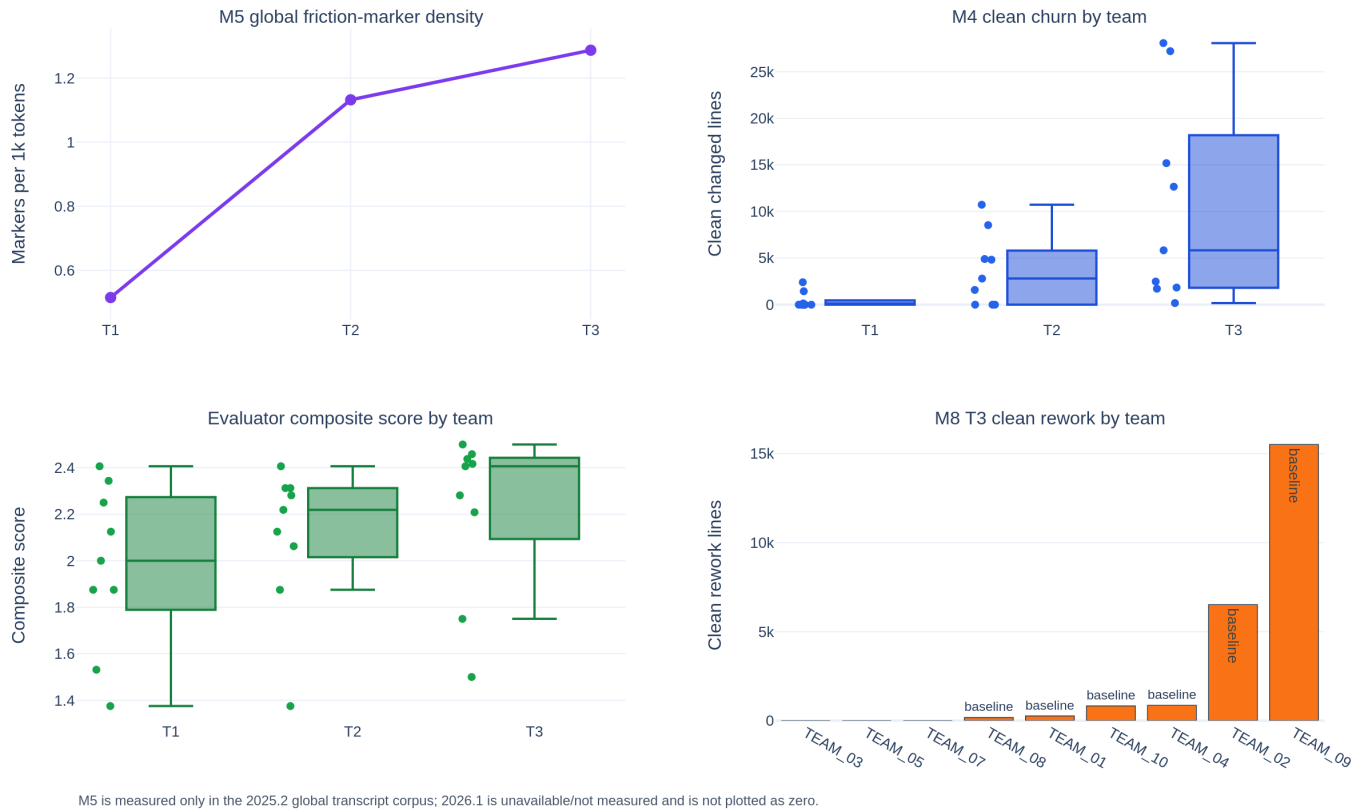


Fig. 11. Coverage-aware M5 triangulation for 2025.2. M5 coordination-marker density is measured in the global transcript corpus, while clean churn and evaluator scores are summarized across the nine 2025.2 team-semesters; 2026.1 is unavailable, not zero-filled.

RQ2 checkpoint summaries with explicit units

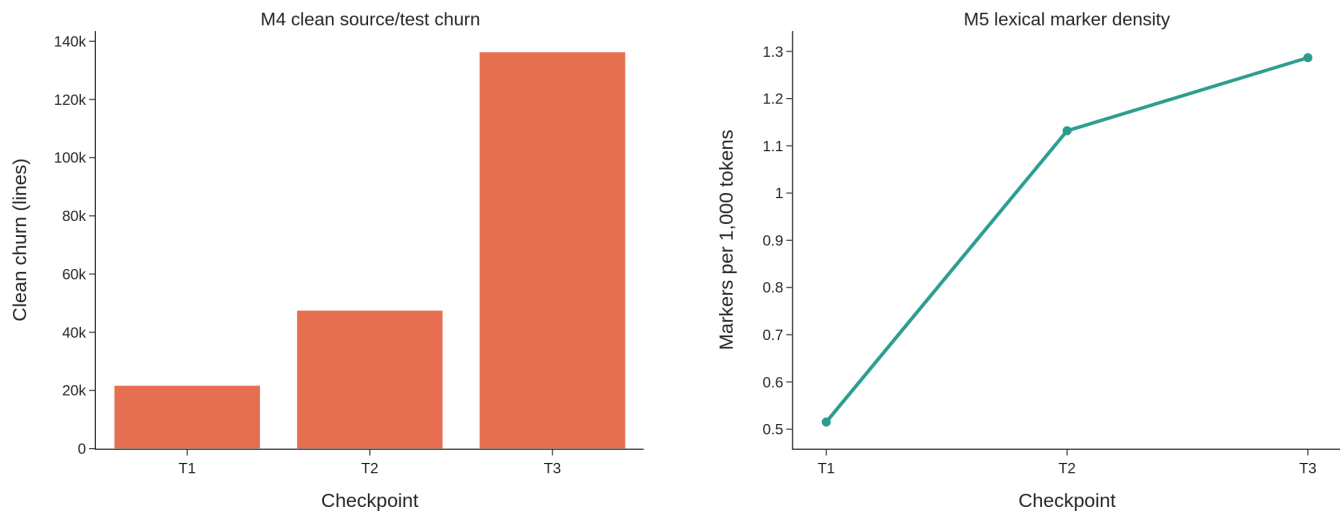


Fig. 12. RQ2 checkpoint summaries with explicit units. Clean-Change Dynamics (M4) and Coordination-Language Evidence (M5) use separate panels and scales.

C. RQ3: How does the observable scope and content evidence of upfront architectural artifacts at baseline T_1 associate with late-stage clean-path rework and independent, blind-scored project outcomes?

RQ3 is evaluated at team-semester grain across only 14 observations. This small sample substantially limits statistical power and the stability of any correlation estimate; accordingly, the reported Spearman coefficients are strictly exploratory diagnostics of structural variation, not confirmatory tests. This narrower analysis complements the broader descriptive evidence for RQ1 and RQ2, which is situated in a cohort of approximately 70 participating students and draws on repeated surveys, repositories from all 14 projects, and the available transcript corpus. These evidence streams retain their distinct analytical grains and are not treated as 70 independent observations. The RQ3 analysis separates observable upfront architectural artifacts from their semantic interpretation, separates rework magnitude from a baseline-conditioned ratio, and treats evaluator outcomes as independent T3 scorecard summaries.

Architectural Artifact Scope (M6a) records whether a T1 architectural artifact was observed and its repository-visible structural scope. An artifact is present in 4 of 9 team-semesters in 2025.2 and in all 5 team-semesters in 2026.1. Among the 2025.2 observations, the median T1 file count is zero and the median T1 line delta is zero; among the 2026.1 observations, the corresponding medians are 4 files and 114 changed lines. The largest observed T1 line delta is 232 in 2025.2 and 222 in 2026.1. These are structural observations, not measures of architectural quality or planning intent. Figure 13 summarizes the cohort-level presence pattern.

Planning Content Evidence (M6b) adds structured evidence from T1 commit subjects for 9 of the 14 team-semesters; 5 cases have no T1 subjects and were not measured. The reviewed fields cover planning evidence, goals, architecture/design, task decomposition, risk/dependency, and literal evidence quotes. They are retained as separate exploratory fields, with no composite planning or architectural-quality score.

Repository Inactivity (M7) shows that repository inactivity is not equivalent to architectural-artifact absence. At T_1 , 7 of 9 team-semesters are inactive in the 2025.2 seven-day window and 3 of 5 are inactive in 2026.1; by T_3 , no team-semester is inactive in either cohort. Repository Inactivity (M7) is therefore used as a repository timing and coverage diagnostic, not as a planning-omission measure and not as a predictor in Artifact–Rework–Outcome Associations (M9).

Clean-Path Rework (M8) reports late-stage changes to previously observed clean paths as a file-provenance proxy for destructive rework. Among the 14 team-semesters, 12 are baseline-eligible. The T3 clean-rework ratio ranges from 0.000 to 1.000 among the reported observations, while clean rework magnitude ranges from zero to 15,518 lines. For example, the largest magnitude occurs in 2025.2/TEAM_09, whereas 2025.2/TEAM_08 and 2026.1/TEAM_03 have a ratio of 1.000

but very different absolute magnitudes (172 and 196 lines). Baseline-ineligible cases remain separate and are not interpreted as stable or low-rework cases. Because path provenance does not establish semantic defects, Clean-Path Rework (M8) is not a direct defect measure. Figure 14 displays the artifact-scope/rework profile on a logarithmic rework axis, preserving baseline eligibility.

Technical complexity is a concurrent descriptive explanation for rework and late concentration. The median T3 technical-complexity evaluator score is 1.875 in the 14 team-semesters. When associations are stratified by this median, planning scope retains a positive association with clean-rework magnitude in both strata ($\rho = 0.45$ in the lower-complexity stratum and $\rho = 0.26$ in the higher-complexity stratum), but several other directions change after stratification. For example, final-seven-day clean-churn concentration versus clean-rework magnitude changes from $\rho = -0.03$ in the unstratified summary to $\rho = -0.47$ in lower-complexity cases and $\rho = 0.54$ in higher-complexity cases. These splits are descriptive small-n sensitivity checks, not regression adjustment. Figures 15 and 16 visualize the complexity profiles used for this alternative explanation.

The primary Artifact–Rework–Outcome Associations (M9) are based on only 14 team-semesters, which sharply limits statistical power and the stability of correlation estimates. We therefore use the coefficients and their p-values strictly as exploratory diagnostics of structural variation, not as confirmatory tests. This narrow inferential base should be distinguished from the broader descriptive evidence used for RQ1 and RQ2, which was collected from the approximately 70 participating students through surveys and their team activity, while retaining the stated grain and coverage limits of each evidence stream. Within these constraints, M9 does not support a general monotonic relationship between T1 artifact scope and later rework. Across all 14 team-semesters, Spearman’s ρ is 0.206 for T1 file count versus T3 clean-rework magnitude and 0.301 for log-transformed T1 line scope versus that magnitude. Within the 12 baseline-eligible cases, the corresponding associations with the T3 clean-rework ratio are -0.032 and -0.060 . Their exploratory p-values range from 0.296 to 0.922.

The independent blind-scored T3 outcomes are similarly weakly associated with the structural predictors. For T1 file count, the correlations range from -0.079 for engagement/participation to 0.271 for technical complexity; for log-transformed T1 line scope, they range from -0.143 to 0.222 across the four outcomes. The outcome summaries themselves vary across the 14 team-semesters: mean project progress is 3.55 ± 0.67 , while the other three recorded outcome means range from 1.76 ± 0.24 to 1.88 ± 0.25 on their respective scorecard scales.

The approved Planning Content Evidence (M6b) fields produce a separate, smaller exploratory view with $n = 9$. Its largest absolute association with clean-rework magnitude is $\rho = -0.408$ for task-decomposition evidence; its largest positive association with an evaluator outcome is $\rho = 0.479$ for task-decomposition evidence versus scope applicability.

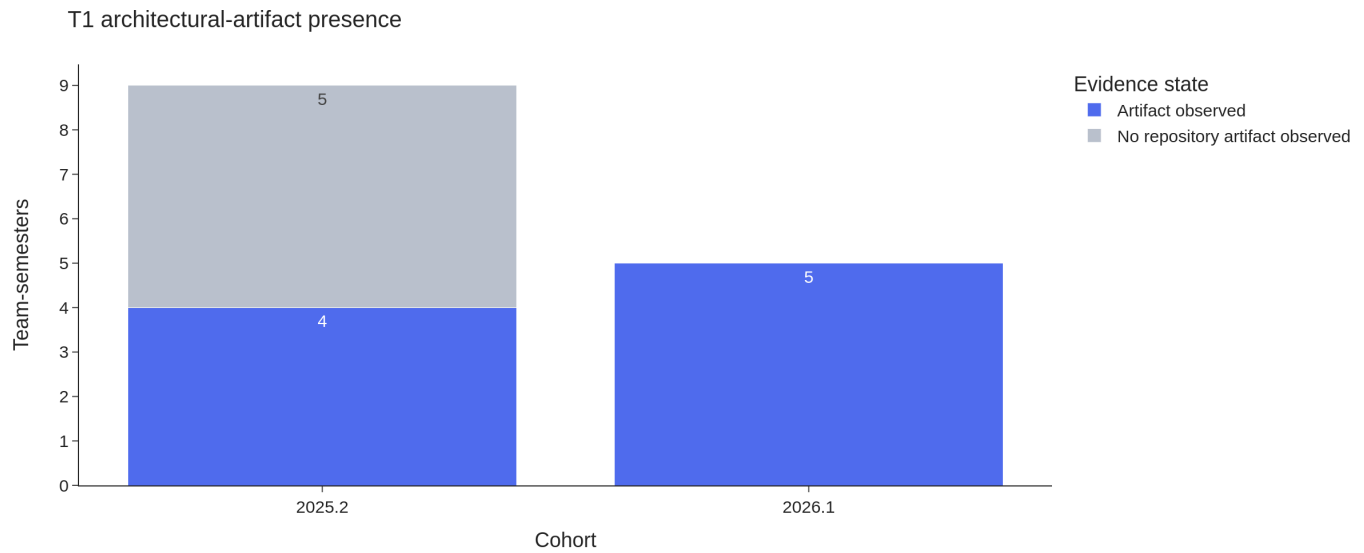


Fig. 13. T1 architectural-artifact presence by cohort. “No repository artifact observed” is an observation state, not evidence that planning did not occur elsewhere.

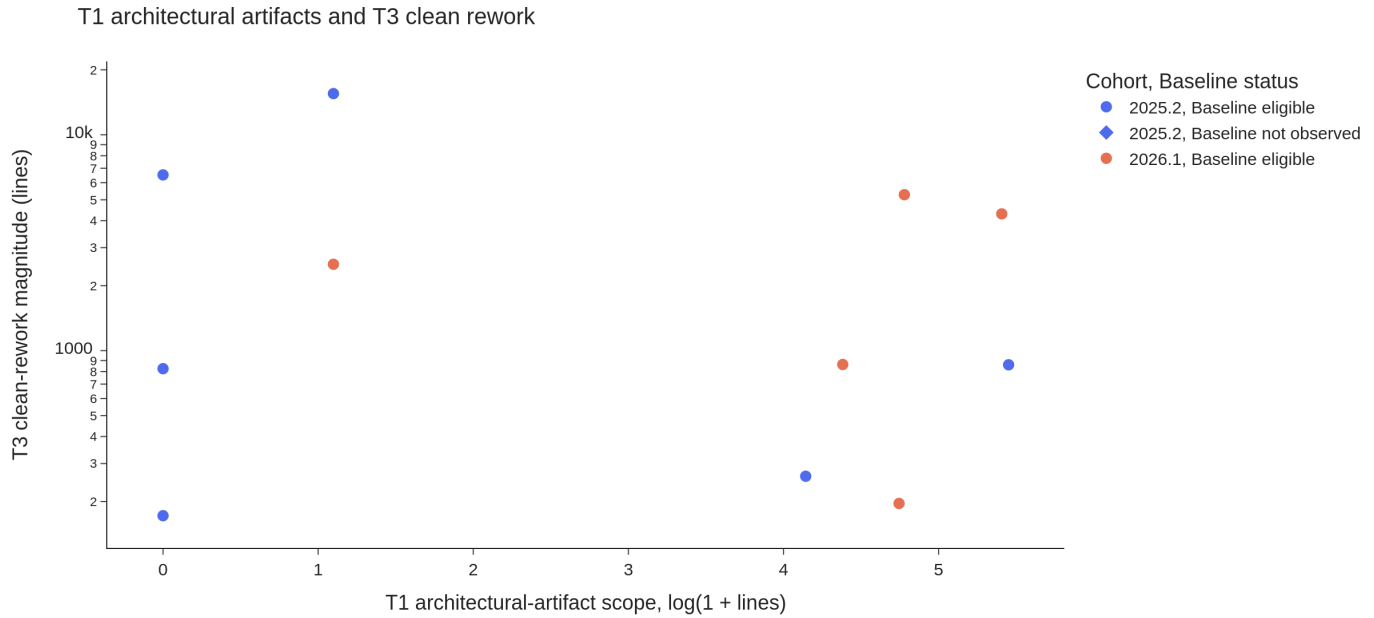


Fig. 14. T1 architectural-artifact scope versus T3 clean-rework magnitude. Points represent team-semester; the vertical scale is logarithmic. Clean rework is a file-provenance proxy, not semantic defect validation.

Leave-one-out diagnostics for Architectural Artifact Scope (M6a) rework analyses produce ranges of 0.078–0.377 for T1 file count and 0.194–0.494 for T1 line scope when predicting rework magnitude. For the eligible-stratum rework ratio, the corresponding ranges are -0.222 – 0.165 and -0.221 – 0.137 . These diagnostics expose sensitivity to individual team-semester rather than confidence intervals. Table VII reports the primary Artifact–Rework–Outcome Associations (M9) values; Figure 17 visualizes the leave-one-out ranges.

Figure 18 extends the leave-one-out view beyond the pri-

mary M9 planning-scope rows to the registered RQ2/RQ3 robustness relationships. Direction is preserved in the full leave-one-out set for several relationships, including planning scope versus final-seven-day concentration, final-seven-day concentration versus T3 score, regularity index versus score delta, and technical complexity versus clean-rework churn. The baseline-eligible technical-complexity–rework-ratio relation is less stable, with sign preservation in 10 of 14 leave-one-out rows. These values are robustness diagnostics for a 14 team-semester panel, not confidence intervals or confirmatory

Technical complexity and T3 clean rework

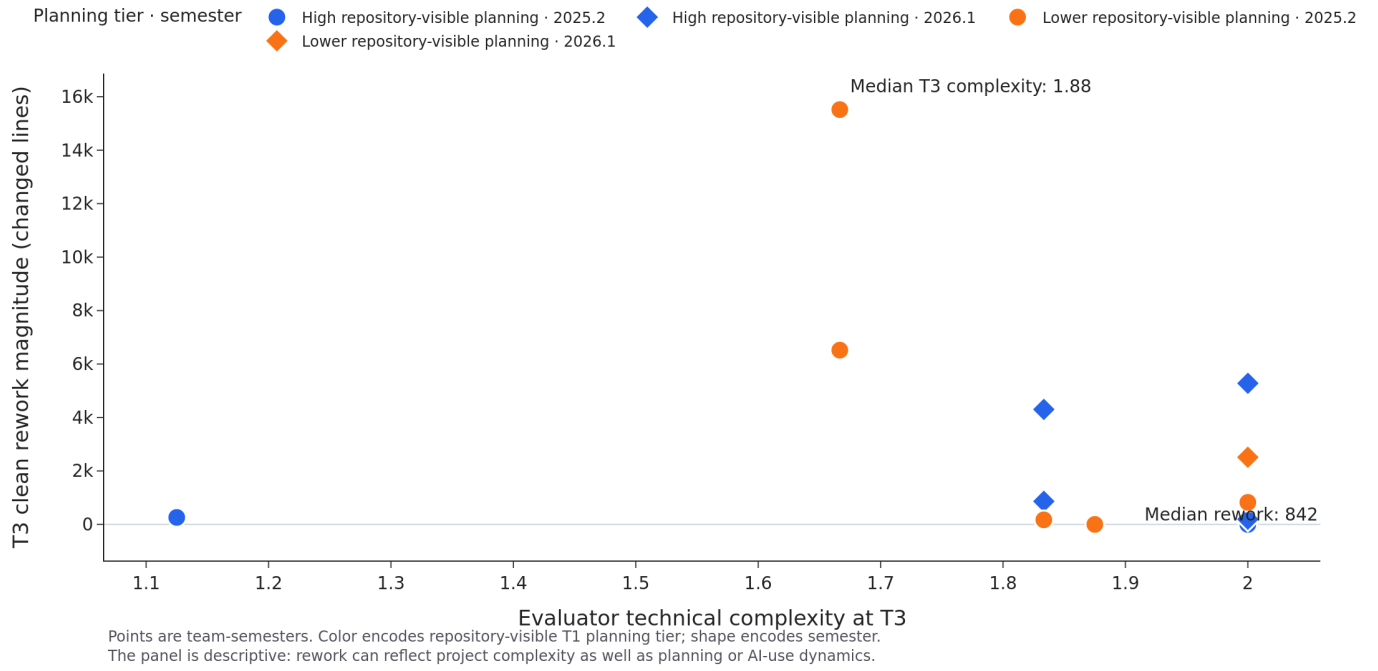


Fig. 15. T3 technical complexity versus clean rework. Technical complexity is an evaluator dimension and clean rework is a clean-path provenance proxy; the figure is descriptive and does not adjust causally for complexity.

TABLE VII
PRIMARY RQ3 ASSOCIATION RESULTS

Predictor	Outcome	Stratum	<i>n</i>	ρ	Exploratory <i>p</i>
T1 artifact file count	T3 rework magnitude	All team-semesters	14	0.206	0.479
T1 artifact line scope	T3 rework magnitude	All team-semesters	14	0.301	0.296
T1 artifact file count	T3 rework ratio	Baseline-eligible	12	-0.032	0.922
T1 artifact line scope	T3 rework ratio	Baseline-eligible	12	-0.060	0.853
T1 artifact file count	Project progress	All team-semesters	14	0.100	0.733
T1 artifact file count	Scope applicability	All team-semesters	14	0.025	0.931
T1 artifact file count	Technical complexity	All team-semesters	14	0.271	0.349
T1 artifact file count	Engagement/participation	All team-semesters	14	-0.079	0.790
T1 artifact line scope	Project progress	All team-semesters	14	0.096	0.745
T1 artifact line scope	Scope applicability	All team-semesters	14	0.038	0.897
T1 artifact line scope	Technical complexity	All team-semesters	14	0.222	0.446
T1 artifact line scope	Engagement/participation	All team-semesters	14	-0.143	0.626

significance tests.

Planning, rework and technical complexity overlay

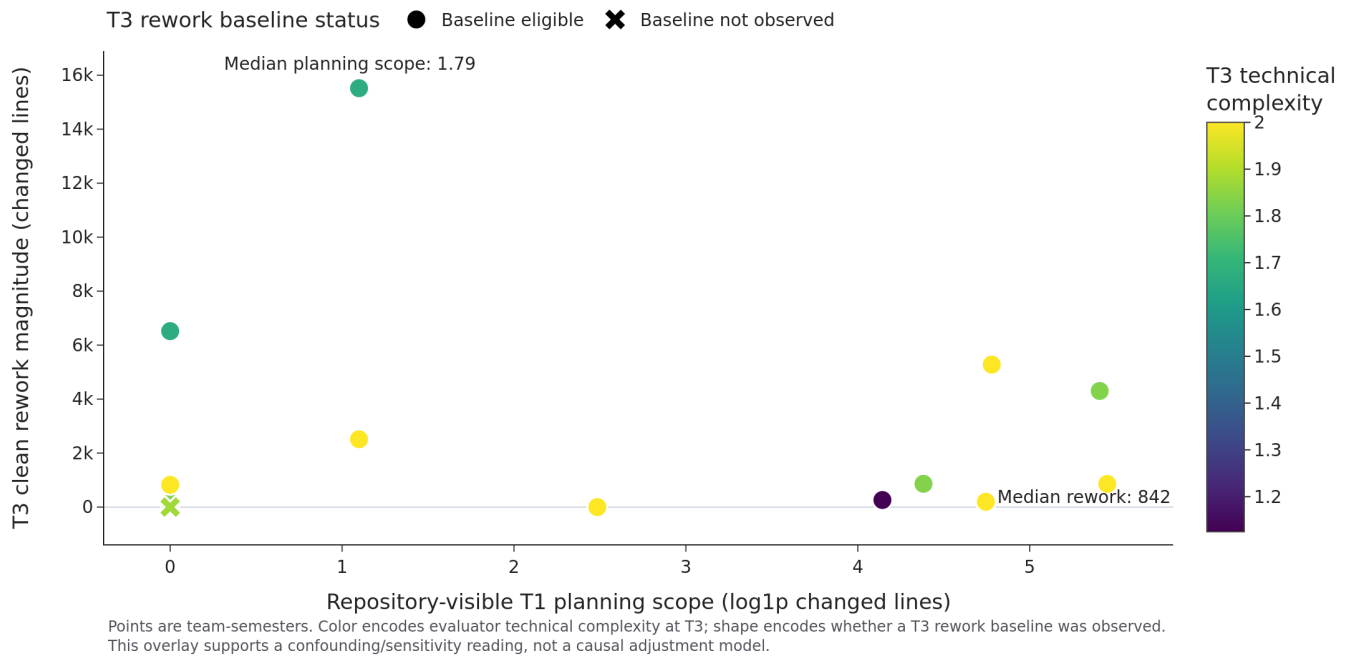


Fig. 16. Planning scope, clean rework, and T3 technical complexity in the RQ3 team-semester panel. The overlay contextualizes rework against technical complexity rather than attributing it solely to planning scope.

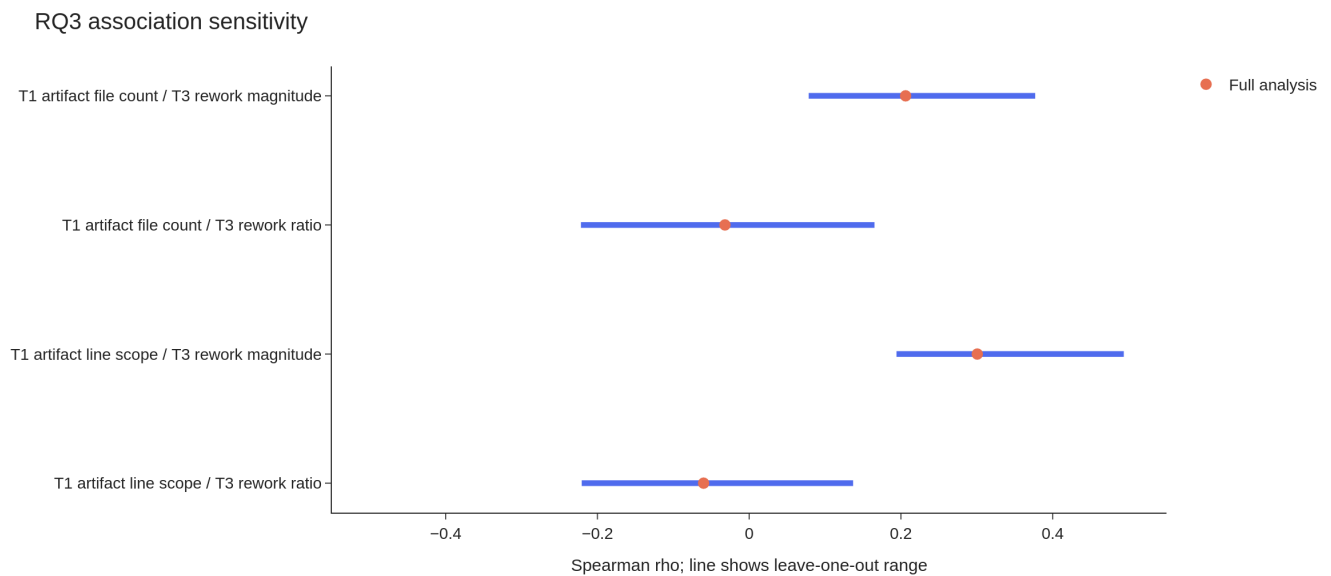
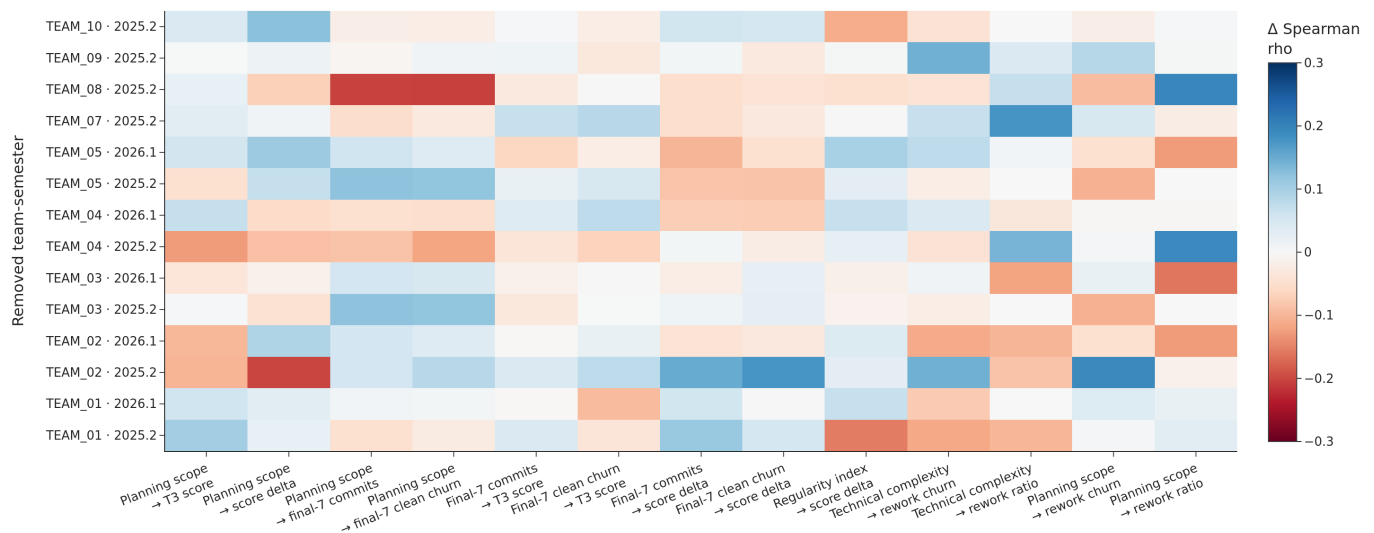


Fig. 17. RQ3 association sensitivity. Points show full-analysis Spearman values; horizontal segments show leave-one-out ranges. These ranges are descriptive diagnostics, not confidence intervals.

Leave-one-out influence map for directional associations



Cells show change in descriptive Spearman rho after removing one team-semester. Use as directional robustness evidence, not confirmatory significance.

Fig. 18. Influence map for registered RQ2/RQ3 relationships. Cells summarize leave-one-out Spearman changes by removed team-semester; the map is a small-sample sensitivity diagnostic rather than a confirmatory test.

Overall, the RQ3 evidence provides a descriptive comparison of observable T1 artifact scope, late clean-path changes, and independent evaluator outcomes, but the 14-team-semester sample cannot support stable confirmatory correlation or establish that artifact quality causes less destructive rework or better project outcomes. The principal result is therefore a bounded, exploratory association profile with explicit missingness and influence diagnostics, not a causal planning-effect estimate.

V. DISCUSSION

The results support a qualified answer to the three research questions. Student developers report frequent use of Generative AI across research, code generation, debugging, documentation, and related activities, but the perception families do not form a single longitudinal trajectory. Repository activity and clean source/test change concentrate near the final evaluation anchor, while candidate coordination-language density increases in the observed transcript corpus. Finally, the study finds observable variation in upfront architectural-artifact scope, but no robust monotonic association with late clean-path rework or independent blind-scored outcomes. The central implication is therefore not that a larger artifact necessarily produces a better project, but that AI-assisted development makes the quality and visibility of shared intent consequential while leaving its effects difficult to identify with small observational samples.

A. From local acceleration to system-level coordination

RQ1 and RQ2 describe two sides of AI-assisted development. Survey responses show that students use Generative AI for concrete development activities, while Repository Activity (M3) and Clean-Change Dynamics (M4) show that repository-visible activity intensifies around final integration. Coordination-Language Evidence (M5) adds textual evidence that candidate coordination markers are more dense at later checkpoints in the observed 2025.2 corpus. These results do not establish that AI use caused coordination friction; the relevant measurement boundaries are consolidated in Threats to Validity. They do, however, weaken a simple productivity story in which faster local code production automatically yields smoother system integration.

The more defensible interpretation is a shift in the bottleneck. Generative AI can reduce the effort required to produce local implementation fragments, but it does not remove the need to agree on architecture, interfaces, task boundaries, dependencies, or acceptance conditions. In a multi-person project, those agreements remain coordination objects even when code is produced more quickly. The late concentration observed in Repository Activity (M3) and Clean-Change Dynamics (M4) is consistent with a process in which integration pressure accumulates until the system must be assembled under deadline, while Coordination-Language Evidence (M5) suggests that coordination-related language is already present in the shared record. This is a process interpretation, not a causal estimate.

The robustness analyses refine that interpretation by showing that the late peak should not be collapsed into a single “student syndrome” explanation. Some team-semesters combine high final-seven-day concentration with a positive T3–T1 score delta, which is consistent with compensatory late execution or productive recovery under deadline pressure. Others combine similarly high late concentration with no evaluator-score gain, which is more consistent with delayed integration that did not translate into assessed improvement. Still others improve with lower final concentration, indicating that distributed execution remains a plausible path to strong outcomes. The point is not that one profile is the true mechanism, but that repository timing, planning visibility, and evaluator trajectories separate several mechanisms that a simple procrastination narrative would conflate.

B. Architectural artifacts: necessary evidence, not a quality score

RQ3 provides an important qualification to strong planning narratives. Architectural Artifact Scope (M6a) measures whether an architectural artifact was observed in the repository and how much structural scope it occupied; Planning Content Evidence (M6b) extracts literal, structured planning-content evidence from T1 commit subjects. Neither metric directly measures architectural quality. Repository absence cannot establish that a team did not plan elsewhere, and a larger artifact cannot establish that its design was coherent or useful.

The M9 associations are therefore modest and unstable rather than uniformly directional. With only 14 team-semesters, RQ3 has low power for confirmatory correlation, and its coefficients serve only as exploratory diagnostics of structural variation. The paper’s stronger empirical contribution lies in the descriptive and triangulated RQ1–RQ2 evidence collected across the approximately 70 participating students, while the survey, repository, and transcript streams retain their distinct grains and coverage. T1 artifact scope has small positive associations with T3 rework magnitude in the complete team-semester set, while the baseline-eligible rework-ratio associations are near zero and slightly negative. Associations with blind-scored outcomes are likewise mixed and small. The leave-one-out ranges show that individual team-semesters can materially change the observed coefficient. The result is not evidence against planning; it is evidence against treating repository-visible artifact volume as a validated measure of planning quality.

The complexity profiles add a second qualification. Late clean-path rework can reflect technical difficulty and project structure as well as planning gaps or AI-assisted implementation dynamics. The stratified summaries show that some associations change direction across lower- and higher-complexity strata, so complexity is best treated as a concurrent explanation that bounds the planning interpretation. This does not make complexity a causal adjustment in the present design; it makes explicit why rework should not be attributed to a single source.

The distinction also matters for interpretation of the tail cases. A team with no observed T1 artifact and a team with

an artifact that is incomplete, misaligned, or unused are not equivalent cases. Conversely, a high rework ratio does not by itself identify a defect: Clean-Path Rework (M8) follows clean-path provenance and cannot determine whether a change was semantically destructive. The empirical chain is consequently useful as an auditable observation system, but not as a single planning-quality scale.

C. Why Spec-Driven Development is a plausible response

The results provide a strong rationale for discussing Spec-Driven Development (SDD). The argument is not that this study tested SDD, nor that SDD is proven to reduce rework. Rather, the observed combination of rapid local generation, multiple late-execution profiles, repeated architectural reformulation in the project records, technical-complexity sensitivity, and persistent coordination demands identifies precisely the class of problem that SDD is designed to address: keeping intent, constraints, architecture, and tasks explicit before and during implementation.

Current SDD guidance describes a workflow in which a specification is carried through technical planning, task decomposition, implementation, and explicit convergence checks [1], [2]. This is aligned with the mechanism suggested by the present evidence. If Generative AI makes implementation cheap while leaving system-level decisions underspecified, a process that externalizes those decisions can reduce ambiguity before it is multiplied across generated code and parallel contributors. The potential benefit is not documentation for its own sake; it is a shared, inspectable context for coordinating people and AI tools.

The claim must remain conditional. The current data do not compare SDD with another process, do not randomize teams or interventions, and do not measure specification adoption as a treatment. SDD is therefore a theoretically grounded intervention proposal for future evaluation, not a causal conclusion of this study. A suitable follow-up would compare teams using an explicit specification-to-plan-to-task workflow with teams using their usual process, while measuring architectural change, integration timing, rework provenance, coordination evidence, and independently scored outcomes.

D. Implications for Agile practice

These findings do not imply that iterative development should be abandoned. They suggest that Agile iteration under Generative AI requires a clearer object of iteration. When code can be generated rapidly, iteration should include shared architectural decisions, executable acceptance conditions, interface contracts, and explicit dependency boundaries, rather than treating working code as the only visible progress signal. This interpretation is compatible with Agile values while challenging a purely code-first implementation rhythm. It also separates adaptive iteration from unexamined delay: late work can be compensatory, neutral, or harmful depending on whether it is connected to shared intent, integration checks, and assessed improvement. Granting teams autonomy to use Scrum, Kanban, other Agile variants, or an unguided approach

was a deliberate feature of the naturalistic course design, preserving the process choices students ordinarily make rather than imposing a common treatment. This choice prevents us from attributing the observed T3 friction specifically to Generative AI or to any one process model; instead, it motivates controlled tests of explicit architectural checkpoints, including SDD-inspired rubrics, within otherwise iterative workflows.

From a pedagogical perspective, the observed coordination bottleneck is ambiguous rather than automatically beneficial. Integration friction may create productive opportunities for students to reason about architecture, interfaces, and dependencies, but it may also reveal only near the deadline that rapid AI-assisted implementation had concealed gaps in that understanding. The present instruments measure activity, perceptions, coordination-language evidence, artifacts, rework proxies, and project outcomes; they do not directly measure individual architectural learning. Accordingly, future classroom studies should pair these process measures with pre/post concept assessments and structured student reflections that can distinguish productive struggle from delayed recognition of foundational gaps.

The practical implication is a stronger planning layer, not a return to a fixed plan that cannot change. Specifications should remain revisable, but their revisions should be explicit, reviewable, and connected to the tasks and implementation they govern. For educators evaluating early SDD artifacts, simple heuristics such as file scope (M6a) are insufficient. Instead, rubrics should assess both shared architectural intent and students' reasoning through explicit API contracts, defined cross-component dependencies, justification of design revisions, and verifiable acceptance criteria before GenAI-assisted implementation begins. That is the specific space in which SDD deserves to be tested: as a way to preserve adaptability while making the intent behind AI-assisted change inspectable and the associated learning visible.

E. Scope of the interpretation

The Discussion is intentionally bounded by the evidence design. RQ1 uses repeated cohort summaries rather than a stable individual panel; the remaining coverage, window, proxy, sample, and exploratory-association boundaries are consolidated in Threats to Validity. These limits make the findings useful for mechanism-oriented hypothesis generation, but not for estimating the causal effect of Generative AI, architectural planning, or SDD on project outcomes. The strongest conclusion is narrower: under AI-assisted development, local generation speed does not eliminate the need for explicit system-level coordination, and SDD is a plausible, testable response to that coordination problem.

VI. THREATS TO VALIDITY

The constructs rely on complementary but imperfect proxies. Self-Reported AI Usage and Perceptions (M1) and Perceived Role Disruption (M2) rely exclusively on self-reported survey measures rather than objective behavioral telemetry. Consequently, we cannot verify exact AI utilization rates,

such as API-call volume or Copilot snippet-acceptance rates, limiting our claims to perceived frequency and reported task orientation rather than technical dependency. Consistent telemetry was not available across the proprietary IDE extensions, hosted services, and student-owned development environments used in this naturalistic bring-your-own-device setting. Multi-checkpoint self-reports were therefore the only common measure of adoption that could be collected without requiring access to vendor-controlled logs or intrusive monitoring of students’ devices. Architectural Artifact Scope (M6a) captures repository-visible T1 architectural artifacts, whereas Planning Content Evidence (M6b) extracts structured evidence from commit subjects. The Planning Content Evidence (M6b) remains exploratory and its categories were subject to human review; neither measure establishes architectural quality or planning intent. Clean-Path Rework (M8) uses file-path provenance to identify changes to previously observed clean paths, not semantic defects or destructive intent. Coordination-Language Evidence (M5) uses deterministic lexical markers in transcript text; its counts are candidate textual evidence, not direct diagnoses of coordination friction. These distinctions constrain every metric’s interpretation and prevent the metrics from being treated as interchangeable measures of quality, effort, or productivity.

The observational design and small sample constrain causal and external validity. In particular, the late-stage concentration of Repository Activity (M3) and Clean-Change Dynamics (M4) remains compatible with ordinary student procrastination (the “student syndrome”). We mitigate this threat by reporting internal contrasts that split final-seven-day concentration by evaluator-score change and repository-visible planning scope. Those contrasts separate cases consistent with compensatory late recovery from cases of late concentration without evaluator-score gain and from cases of more distributed progress. However, without a non-AI control group or timestamped AI-use telemetry, we cannot determine whether the timing patterns reflect GenAI-related architectural debt, conventional deadline pressure, or some combination of both.

Windowing and process heterogeneity are also bounded rather than eliminated. Rolling windows overlap, so adjacent M3 values are not independent observations; we mitigate this by adding mutually exclusive phase bins and final-seven-day phase shares, but the bins still derive from repository-visible events and cannot observe work outside Git. Teams had complete autonomy over their process models, including Scrum, Kanban, other Agile variants, and unguided approaches; the study neither standardized these processes nor measured adherence to them. Operational-regularity proxies partially contextualize this heterogeneity by summarizing observable temporal and author regularity in Git, but they are not validated measures of agile-process quality, productivity, or methodological adherence. We therefore cannot rule out that some integration friction arose from process choice or poor methodological adherence rather than AI-tool use.

Artifact–Rework–Outcome Associations (M9) are exploratory, not causal or confirmatory. The study covers 14

team-semesters in one educational setting, and influence maps show that some relationships are sensitive to individual team-semesters. We report leave-one-out ranges and direction-preservation diagnostics to make this sensitivity visible, but those diagnostics are not confidence intervals and do not solve the small-sample problem. Analytical grains differ across student responses, team-semesters, rolling windows, and a global transcript corpus, so denominators cannot be pooled implicitly. Transcript evidence is available only for 2025.2, which restricts cross-cohort comparability for the lexical findings; the M5 triangulation therefore treats 2026.1 as unavailable rather than as zero evidence.

Git cannot observe planning or collaboration outside the repositories. Five Planning Content Evidence (M6b) cases lacked T1 commit subjects and were not measured; missing evidence was not converted into a low score. The course required an integrated GenAI component but did not mandate retrieval-augmented generation (RAG) or another common architecture; none of the 14 projects implemented a full RAG pipeline. This removes mandatory RAG complexity as a shared explanation for the observed integration friction. Nevertheless, the teams’ heterogeneous, self-selected AI integrations differed in technical complexity. We mitigate this threat by reporting complexity profiles and median-stratified sensitivity summaries, but architectural and implementation complexity remain potential project-level confounds that this observational study cannot isolate.

The external SDD literature therefore motivates a testable intervention hypothesis, but this study neither implements nor evaluates SDD. Its results support the need to test more explicit specification and coordination practices under Generative AI, not a claim that SDD has already improved outcomes.

VII. CONCLUSION

Do we still need to be Agile? Yes, but Generative AI changes what effective agility requires. Iteration, feedback, and adaptation remain valuable; however, faster local implementation does not remove the need for shared architectural intent, explicit interfaces, dependency boundaries, and reviewable acceptance conditions. Across the three research questions, this study shows that self-reported AI use is widespread and task-oriented, repository activity concentrates near final integration, and observable upfront architectural artifacts do not translate into a simple monotonic pattern of rework or project outcomes. The late activity peak is not a single phenomenon: the internal contrasts distinguish cases consistent with compensatory late recovery, cases of late concentration without evaluator-score gain, and cases of more distributed progress.

The study’s contribution is to make this tension empirically inspectable through complementary evidence: longitudinal perception and adoption reports, temporal repository measures, transcript-based coordination evidence, structural T1 artifacts, path-based rework measures, and independent evaluator scorecards. The robustness checks do not remove the observational limits of the study, but they make the main validity threats inspectable through non-overlapping phase

bins, internal timing/outcome contrasts, operational-regularity proxies, influence diagnostics, complexity profiles, and explicit transcript-coverage boundaries. Together, these results suggest that GenAI-assisted development shifts a larger share of engineering difficulty away from producing local code and toward coordinating system-level intent. The practical implication is not a return to immutable Big Design Up Front, but a stronger planning layer that remains iterative, explicit, and shared by people and AI tools.

Spec-Driven Development is a plausible way to instantiate that planning layer: requirements become specifications, specifications become technical plans and tasks, and implementation is repeatedly checked against those artifacts. This study does not evaluate SDD as an intervention, so its value here is a grounded and testable proposition rather than an established causal conclusion. The case for SDD is that it directly targets the coordination gap made visible by the combined timing, planning, rework, complexity, and influence diagnostics. Future work should compare ordinary practice with an iterative SDD condition while measuring specification quality, consented AI-tool use, integration timing, operational regularity, technical-complexity strata, coordination evidence, rework provenance, and independently scored outcomes.

VIII. DATA AVAILABILITY

To support independent replication and comply with Open Science policies, an anonymized replication package is available online [18]. This repository contains the deterministic lexical coding scripts, Git extraction queries, and the anonymized dataset required to reproduce the metric derivations, while protecting participant privacy as outlined in Section III-C. Given the volume, heterogeneity, and relational complexity of the underlying data, the replication package is complemented by a fully interactive web application. The application provides researchers with a user-friendly environment in which to navigate the data, explore the study's different evidence sources and levels of analysis, apply filters, and inspect the measures and results without requiring direct manipulation of the raw data. It is intended both to improve the accessibility and transparency of the evidence and to support more efficient examination of patterns that would be difficult to interpret from static files alone. The web application is available online [19].

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